

UVA DS2003 Communicating with Data

Fall 2026

Perception (part II)

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based on materials by Hannah Bako, Nur Yildirim, Adam Perer, and Zhicheng Liu

Quiz

Surprise! You get a 10/10 if you answer the following two questions (no right answers)...

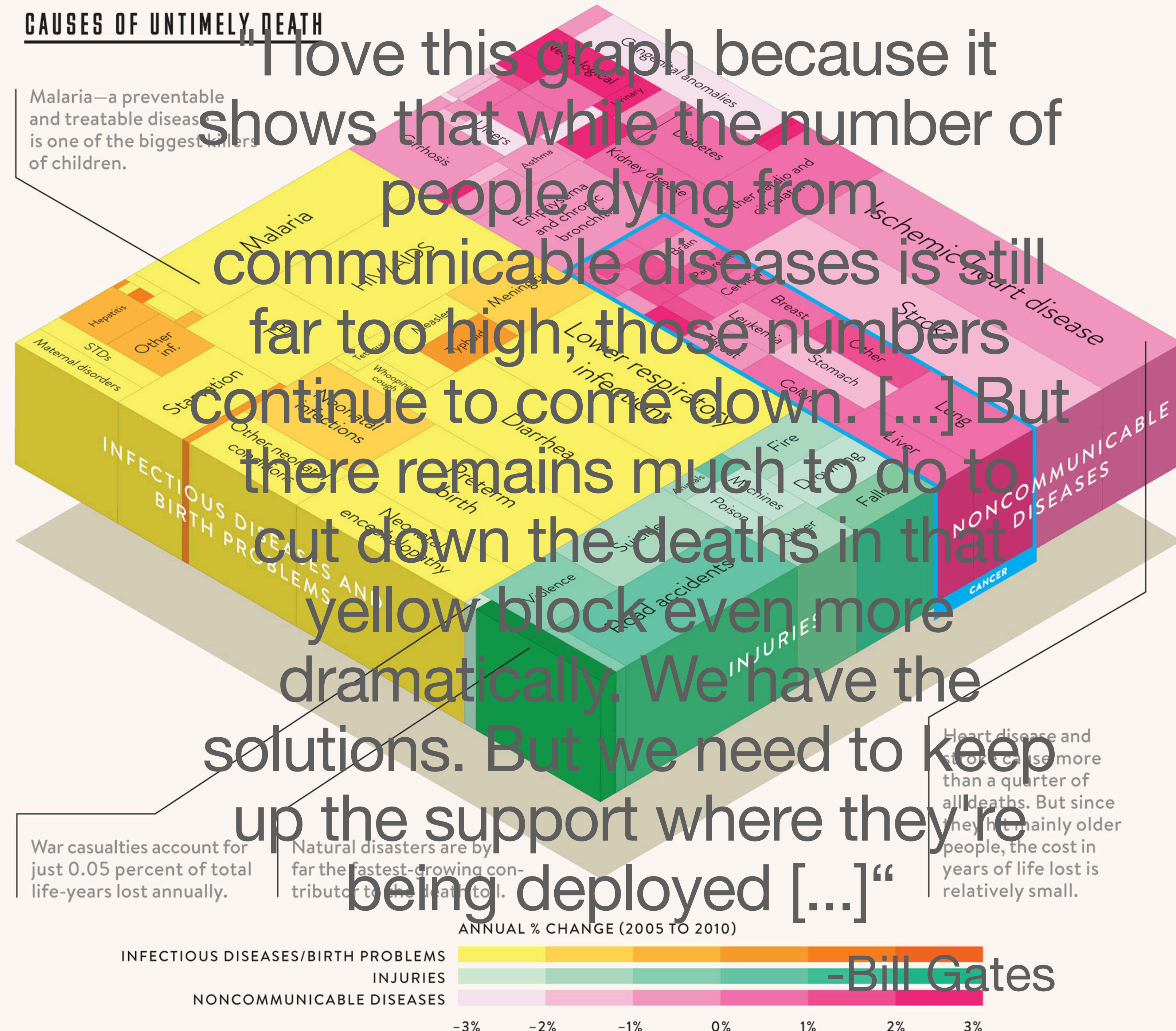
Which chart do you like better?

Which do you think more accurately reflects the data?

A

CAUSES OF UNTIMELY DEATH

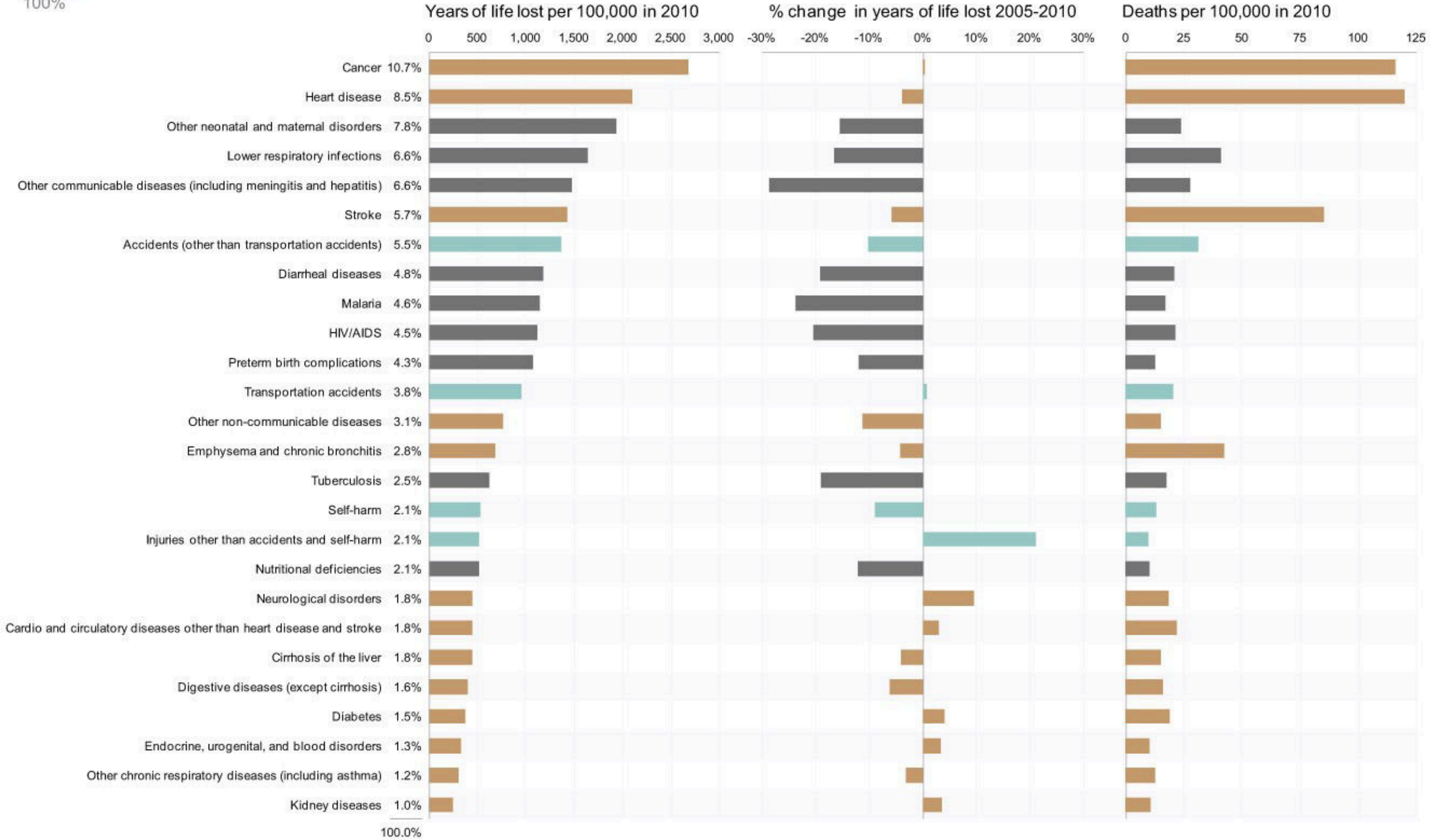
Malaria—a preventable and treatable disease—is one of the biggest killers of children.



B Global Causes of Lost Life

44% Communicable, maternal, neonatal, and nutritional disorders
43% Non-communicable diseases
13% Injuries
100%

Comparing the number of deaths alone, as shown in the rightmost graph below, doesn't tell the entire story. Some causes of death have a greater effect on the young, which can be seen when comparing years of life lost in the leftmost graph.



Some causes of death contribute disproportionately to years of life lost because of their effect on the young. For example, malaria, while not huge in the number of deaths, is much more significant in the number of years that are lost.

Two interesting changes reside in "Injuries other than accidents and self-harm." War, which accounted for only 0.05% of years of life lost, decreased since 2005 by 31.5% in years of life lost per 100,000 people. Natural disasters, which accounted for 0.65% of years of life lost, increased by 217% in years of life lost per 100,000.

Communicable, maternal, neonatal, and nutritional disorders (the gray bars) are often easier to prevent through healthcare than other causes of death. This reveals itself in the graph above by the fact that all of these disorders have decreased during this five year period.

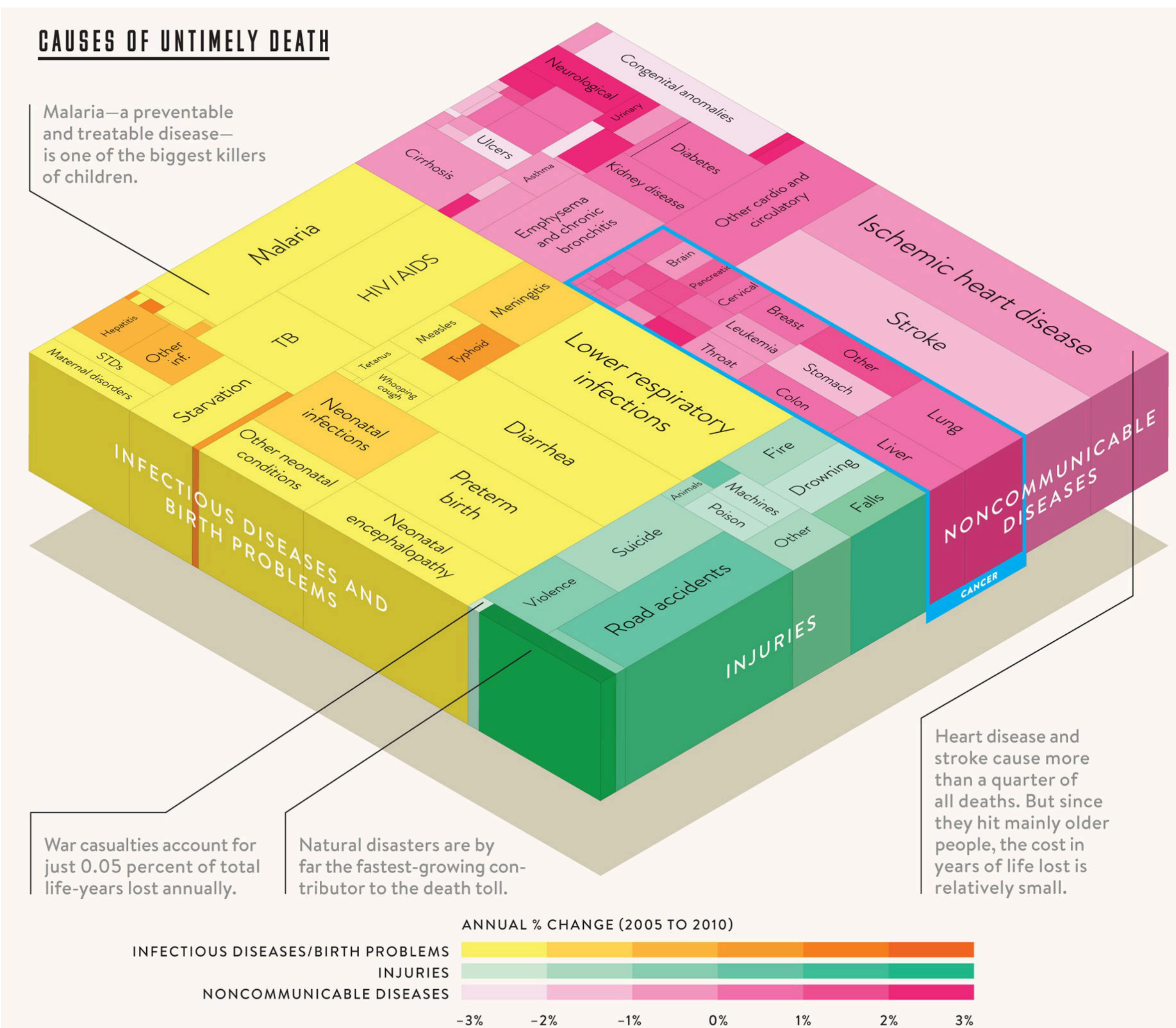
The five forms of cancer that cause the most deaths are trachea/bronchus/lung (2.9%), stomach (1.4%), liver (1.4%), colon/rectum (1.4%), and breast (0.8%).

All cardiovascular and circulatory diseases combined account for 30% of deaths.

Which chart do you like better?

Which do you think more accurately reflects the data?

A



B

Global Causes of Lost Life

Recap

Perception

- Identification and interpretation of sensory information
- From the physical stimulus to recognizing information
- Shaped by learning, memory, expectation
- E.g., hearing someone speak

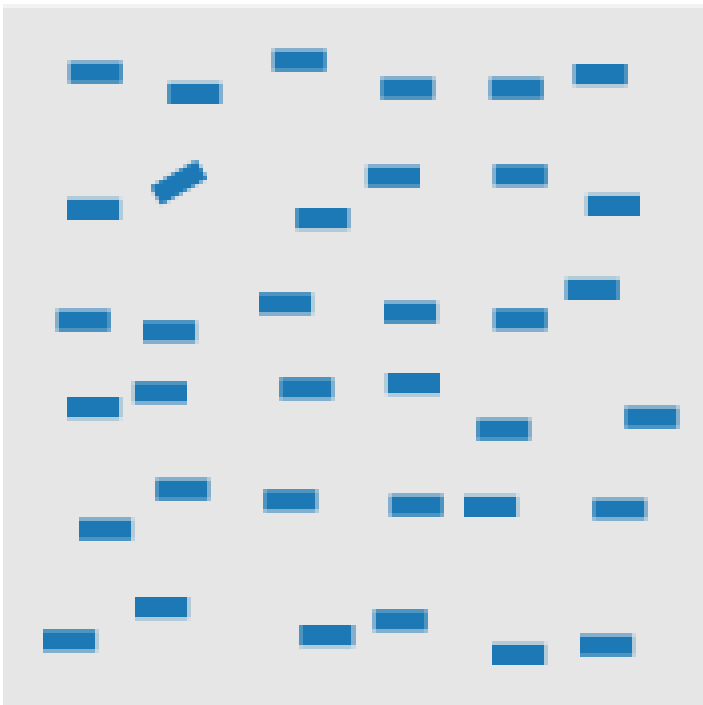
Cognition

- The processing of information, applying knowledge
- E.g., Understanding language and words

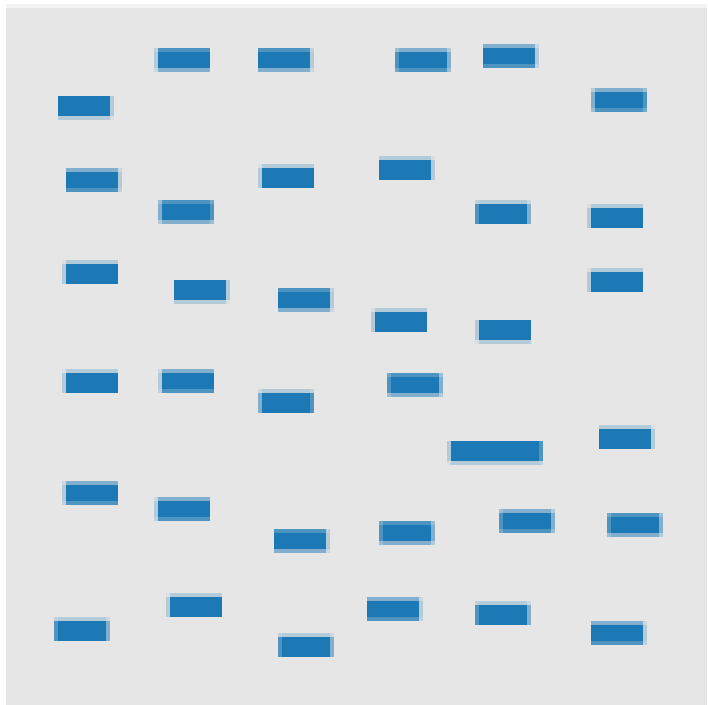
We are talking about what happens **automatically** when a viewer looks at a chart you designed

Recap

Some Pre-attentive Properties



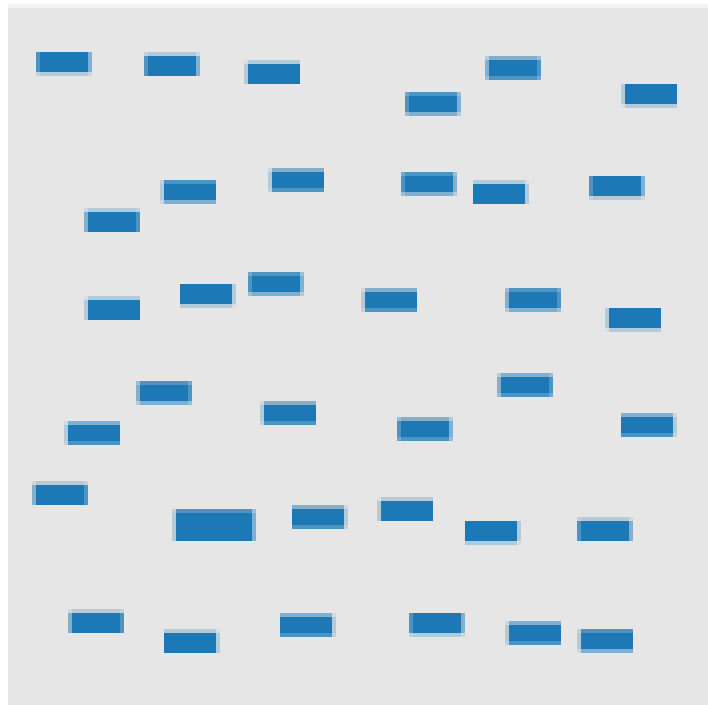
orientation



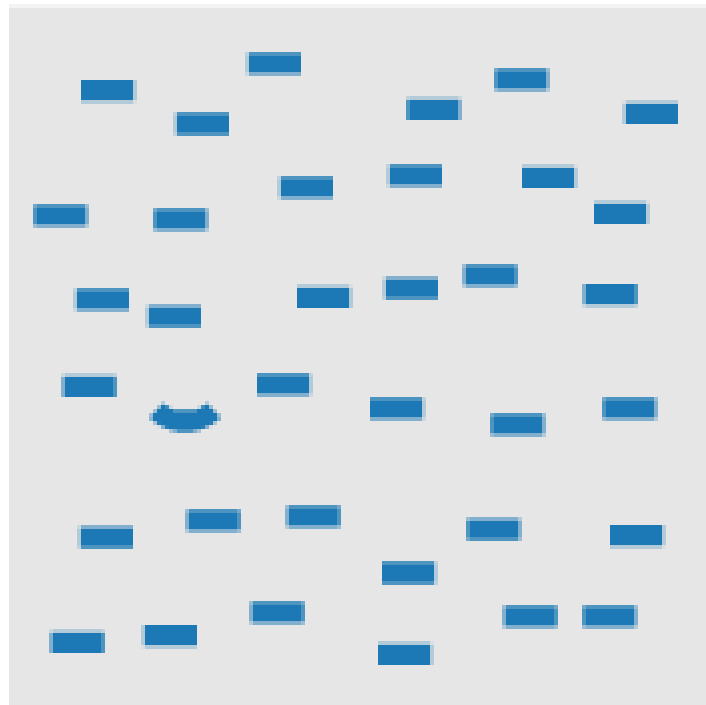
length



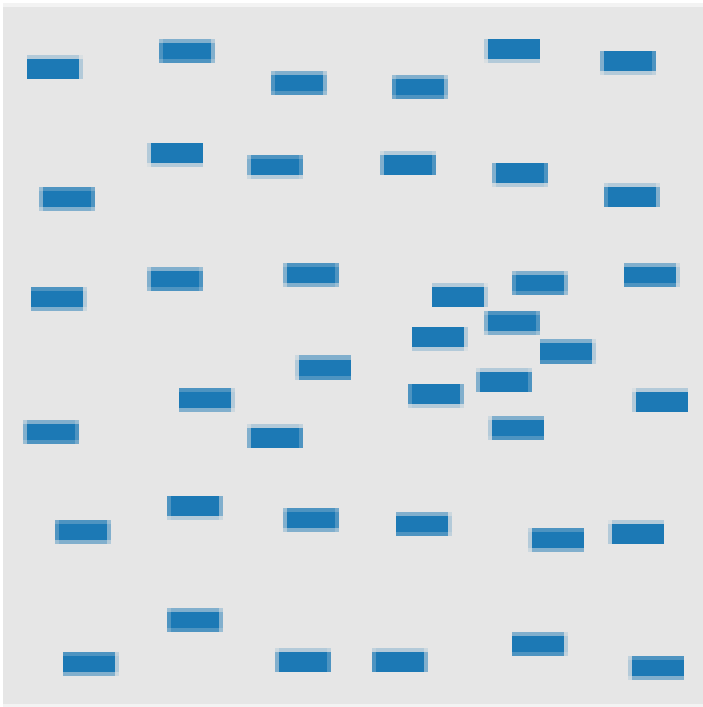
closure



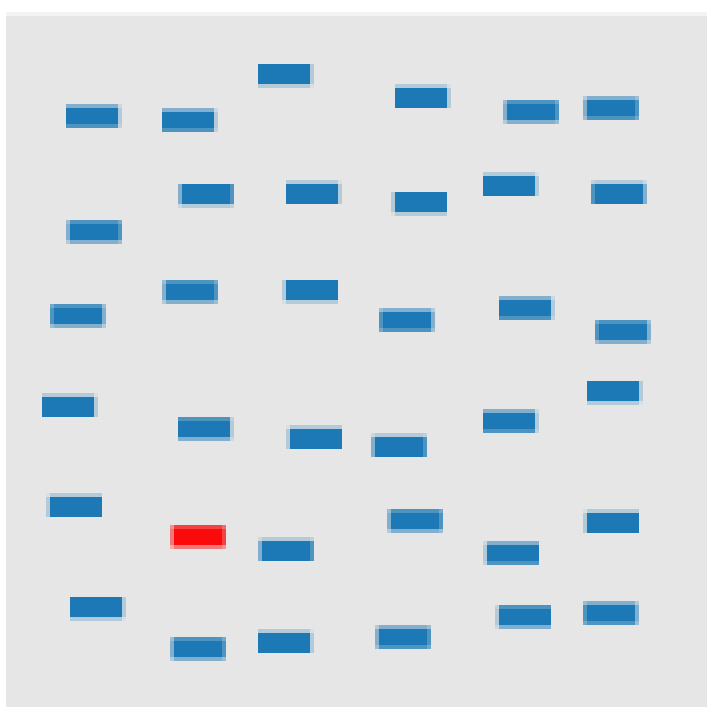
size



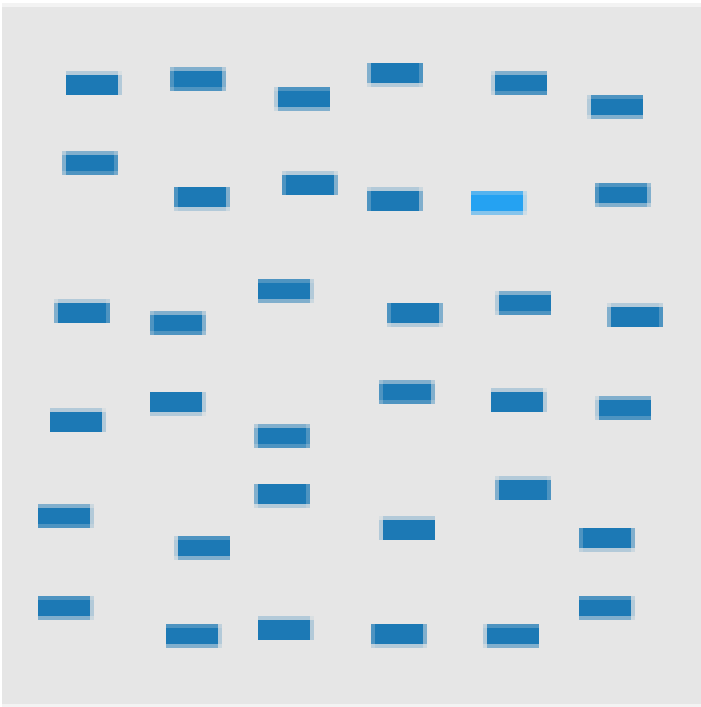
curvature



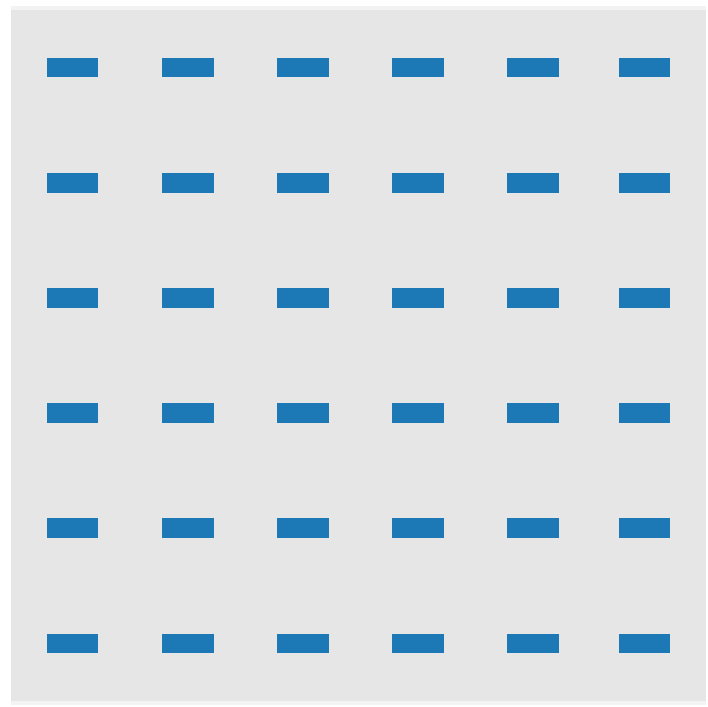
density



hue



hue



flicker



direction of motion

Gestalt Principles

**Expanding on “grouping”
How we see broader patterns**

Gestalt Psychology

Patterns that transcend the visual stimuli that produced them

- Describe how humans perceive visuals in connection with different objects and environments
- “Gestalt” means “pattern” or “configuration” (German origin, 1900s)

We talked about some of these gestalt laws in ways to “group” – proximity, similarity, connectivity

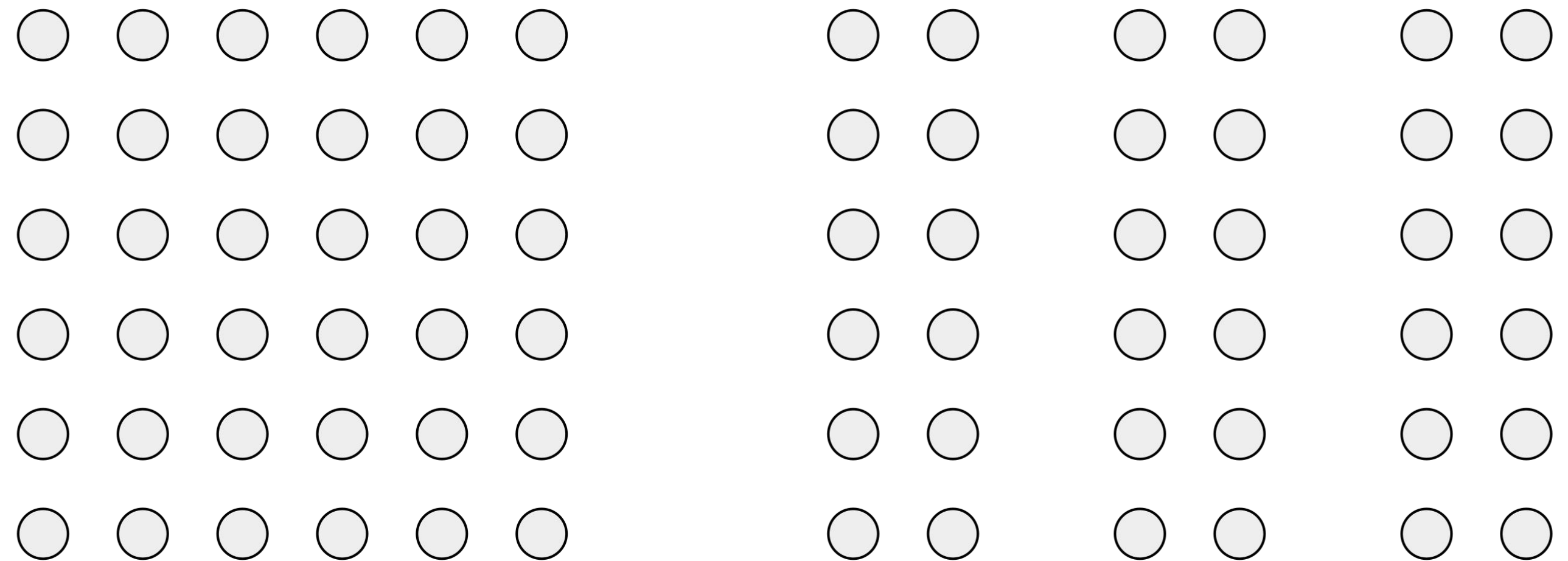
Gestalt Laws

- Proximity
- Similarity
- Symmetry
- Connectedness
- Continuity
- Closure
- Common Fate

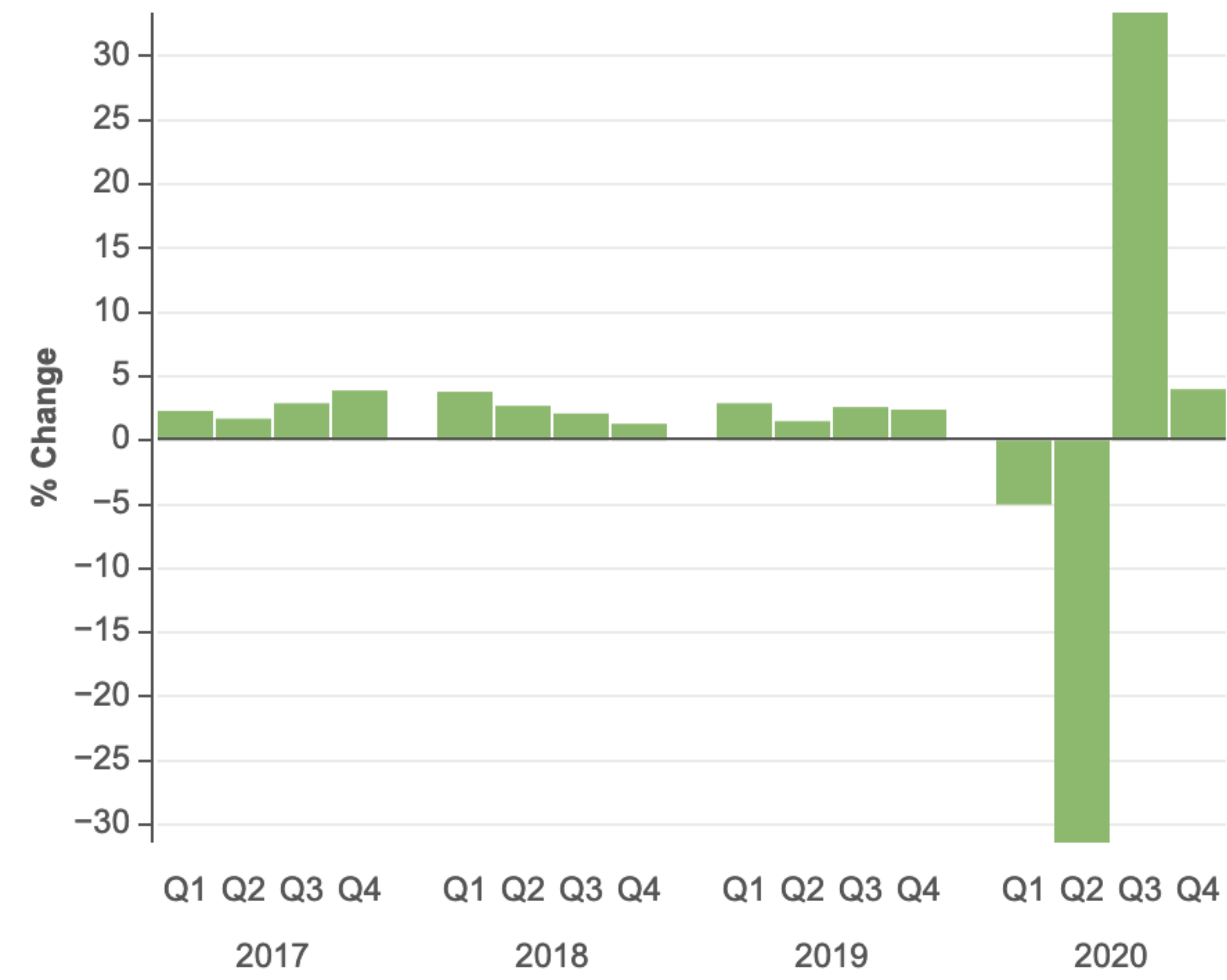
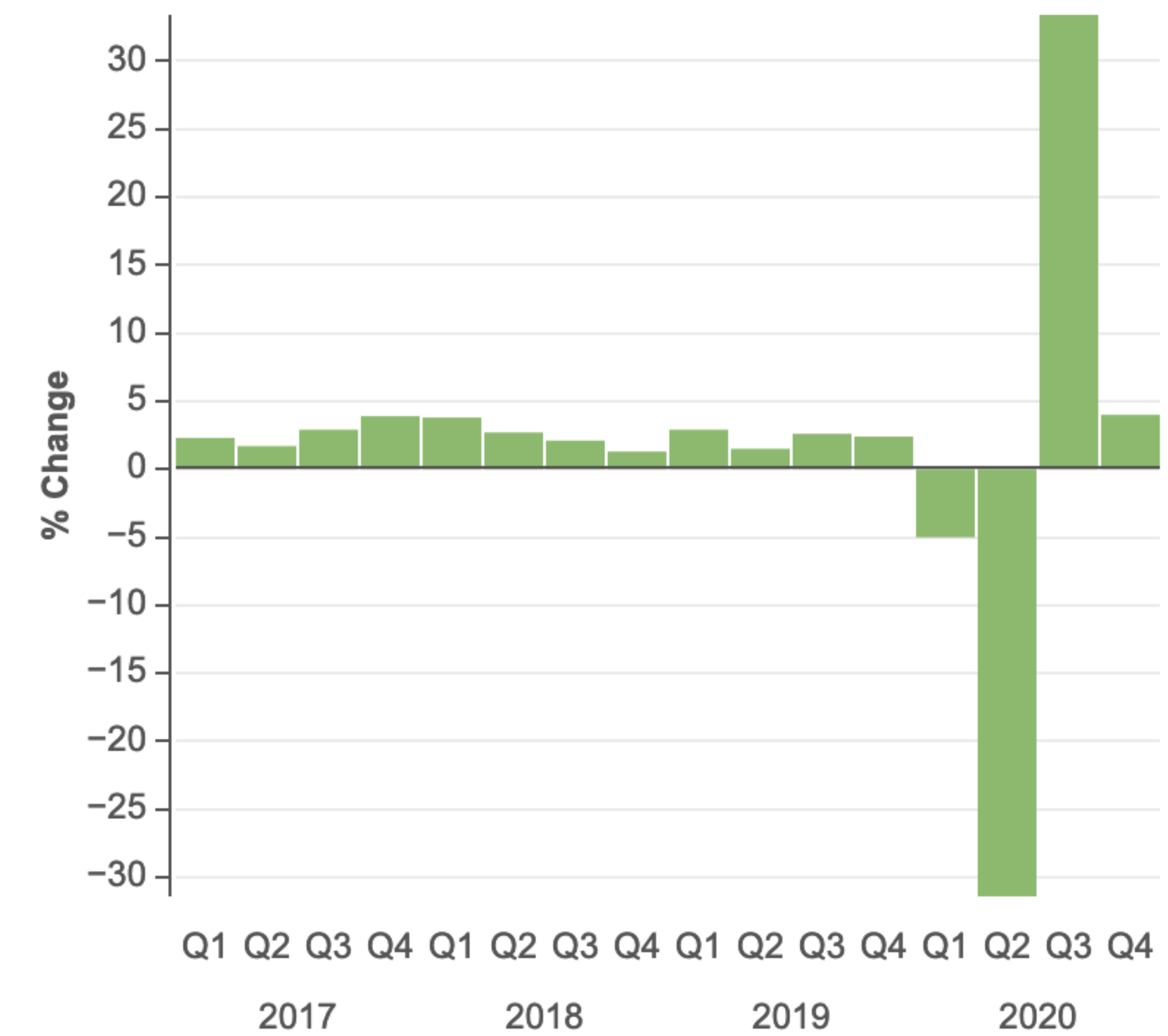
Gestalt Laws

- Proximity
- Similarity
- Symmetry
- Connectedness
- Continuity
- Closure
- Common Fate

Objects that are close to each other are perceived as a group



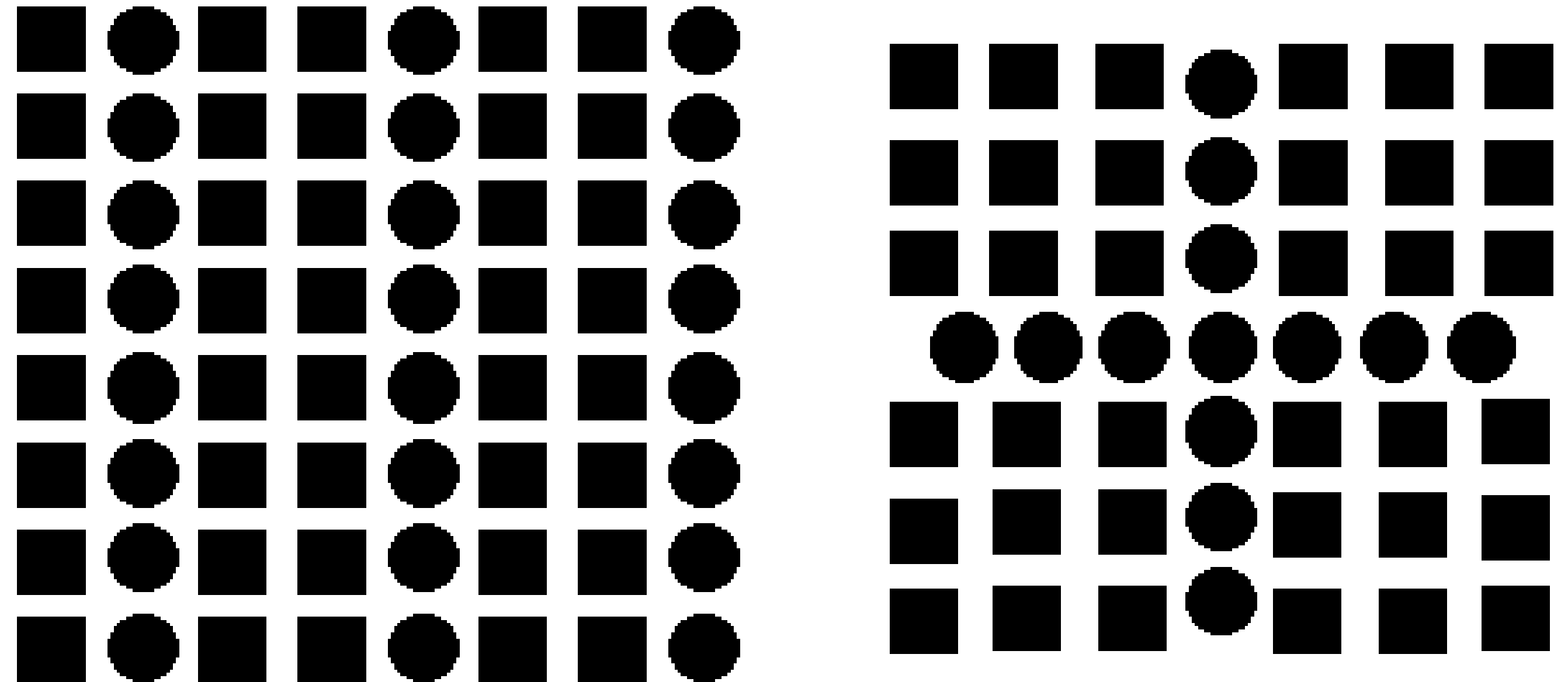
Which of the charts below better utilizes proximity?



Gestalt Laws

- Proximity
- **Similarity**
- Symmetry
- Connectedness
- Continuity
- Closure
- Common Fate

Objects are perceptually grouped together
if they are similar to each other

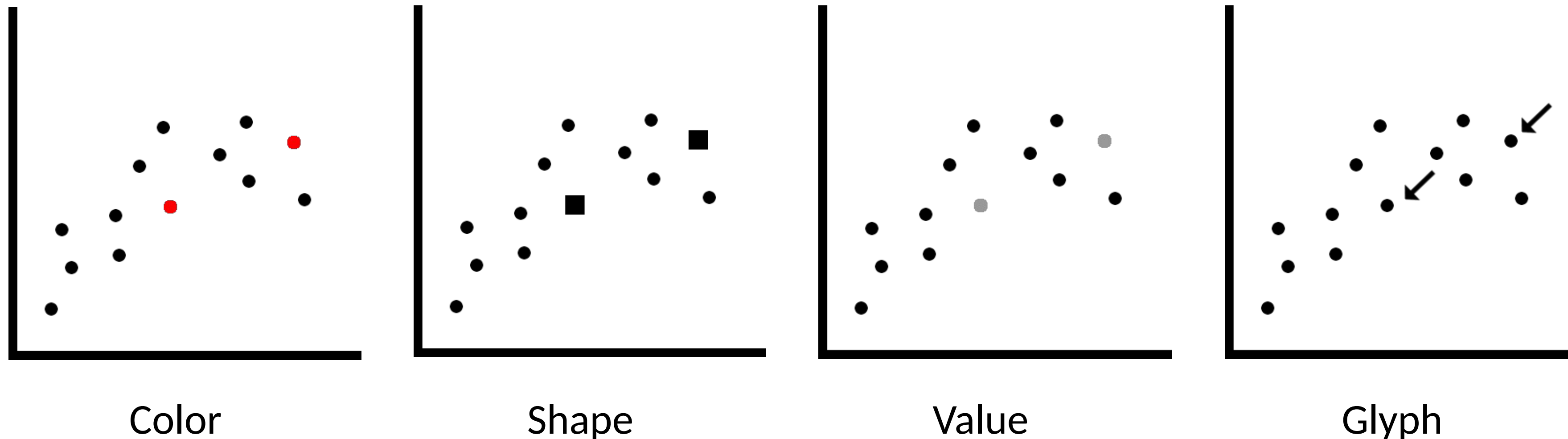


Similarity

Co-modulation of a **channel**

color, shape, size, value, orientation, texture, ...

Adding a glyph, label, frame, background

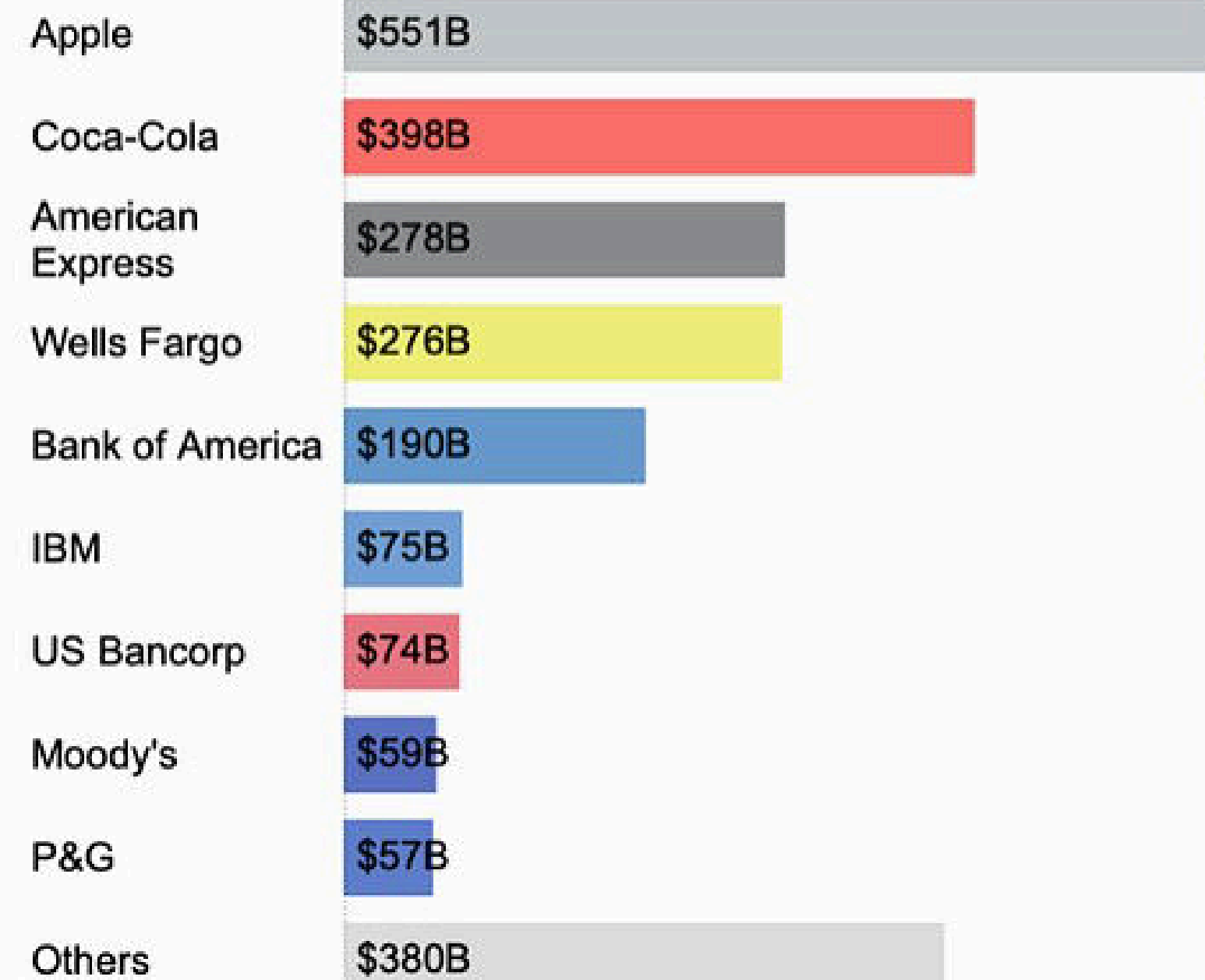


How is color similarity being used across these charts?

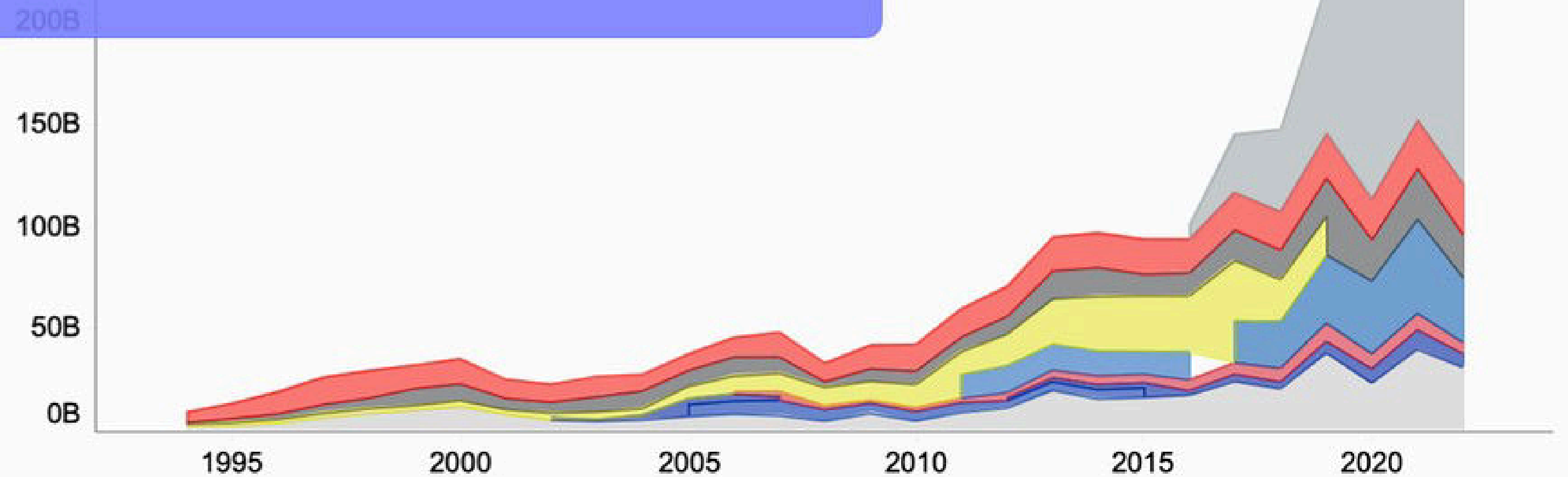
Top-10 Companies by Holdings

For top 20 companies colors are brand based – lots of brands with blue and red colors!

Click to filter and you can also filter the top by year on the right



Different charts but same color for the same categories.
This also helps to utilize charts as a color key.

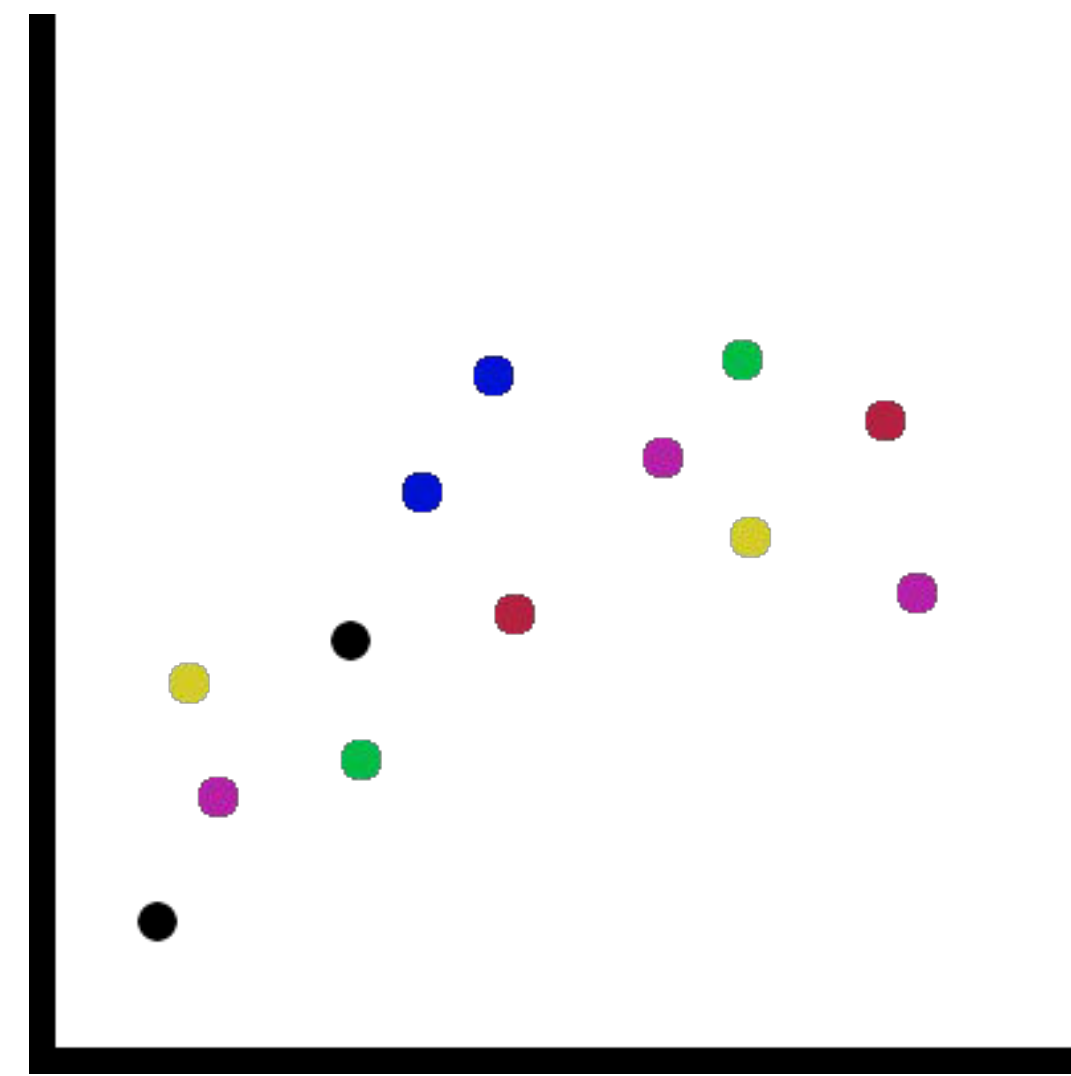
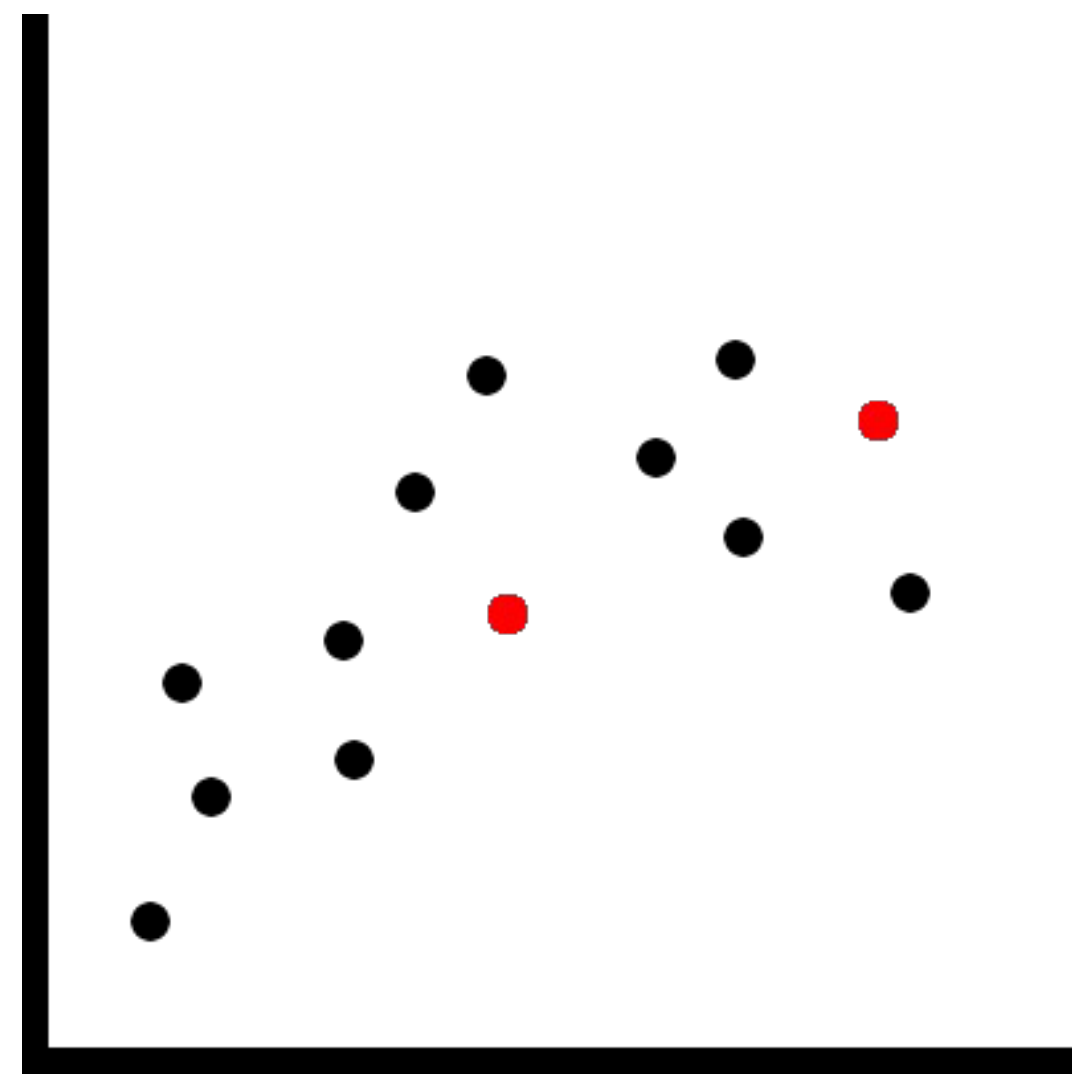


Color – Perception Issues (1/2)

Preattentive properties:

Very good for 1-2 simultaneous

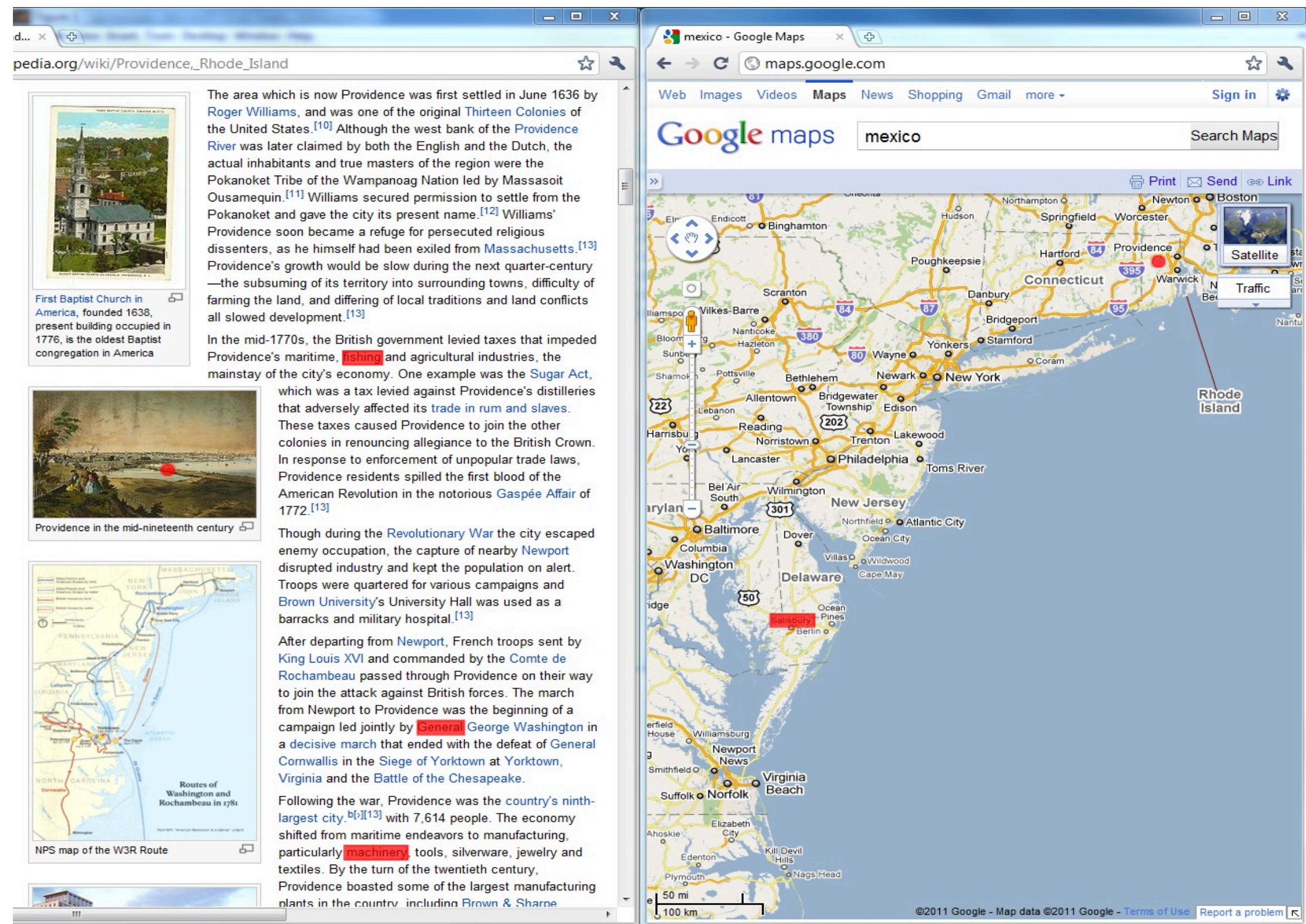
Serial search for more



Color – Perception Issues (2/2)

Slower in a cluttered environment

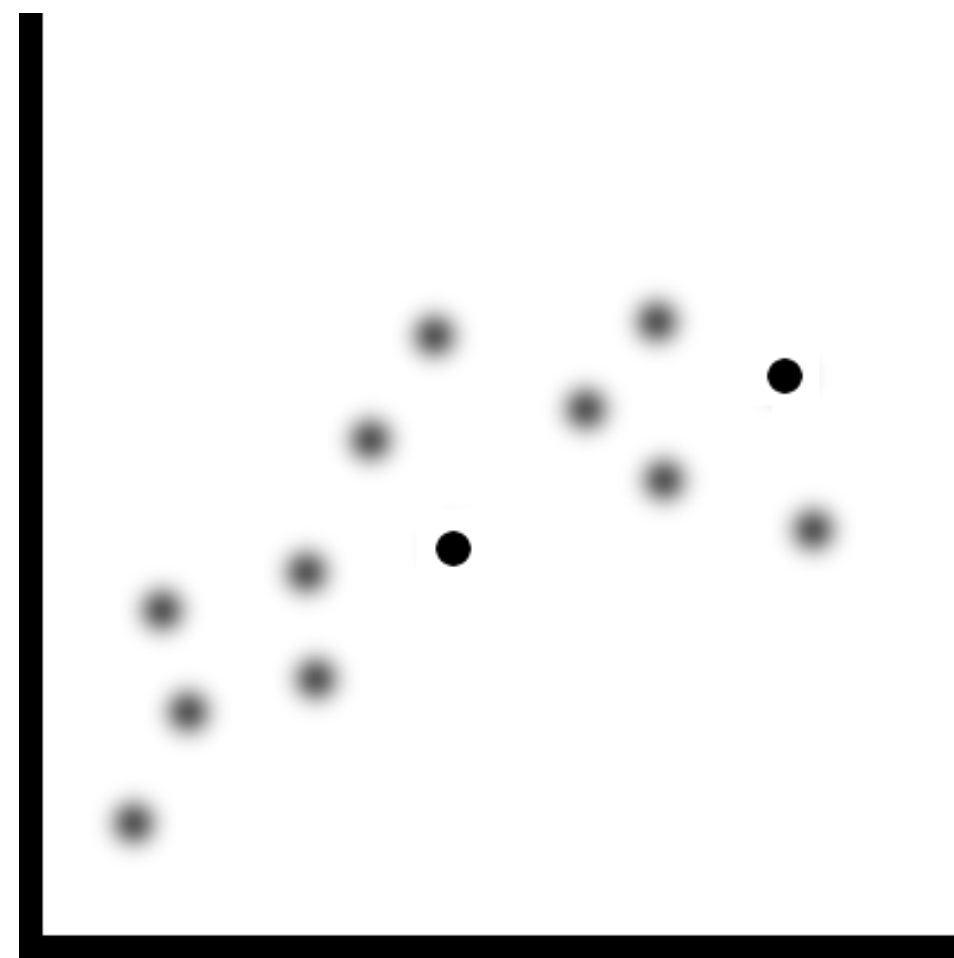
Size of colored object relevant.



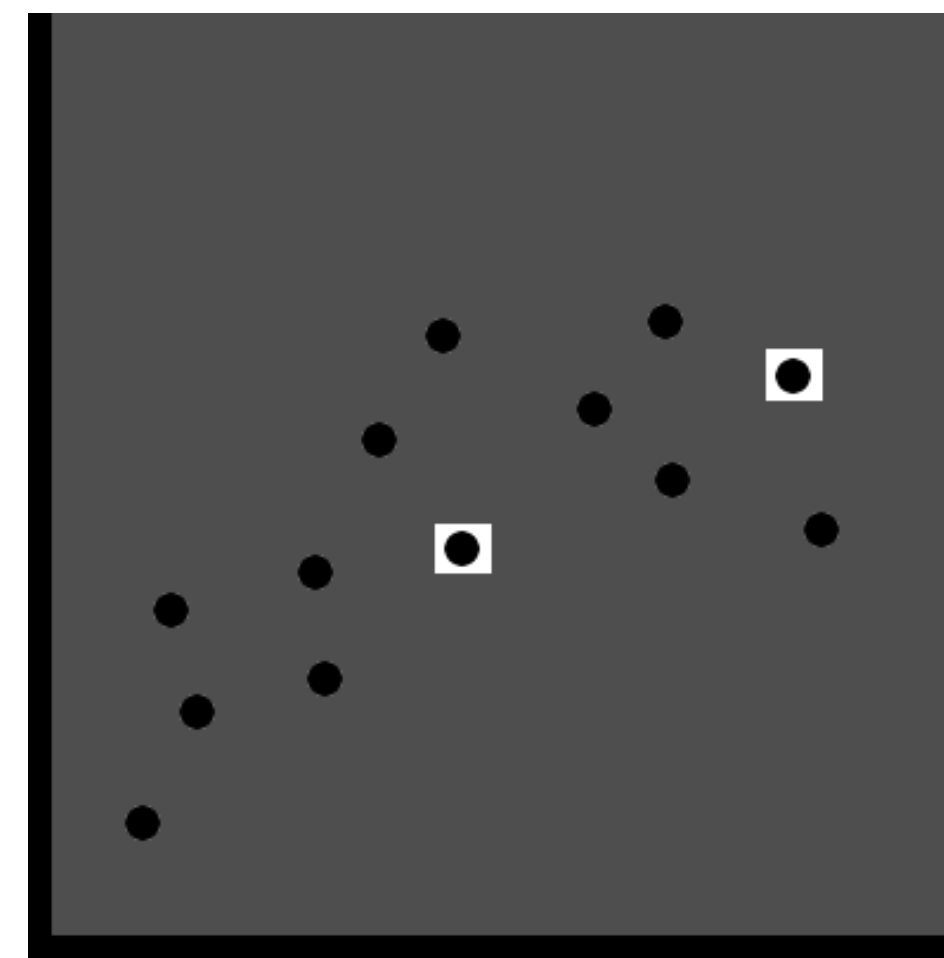
Similarity

Modulate everything else

Blurring, darkening, desaturating, etc.



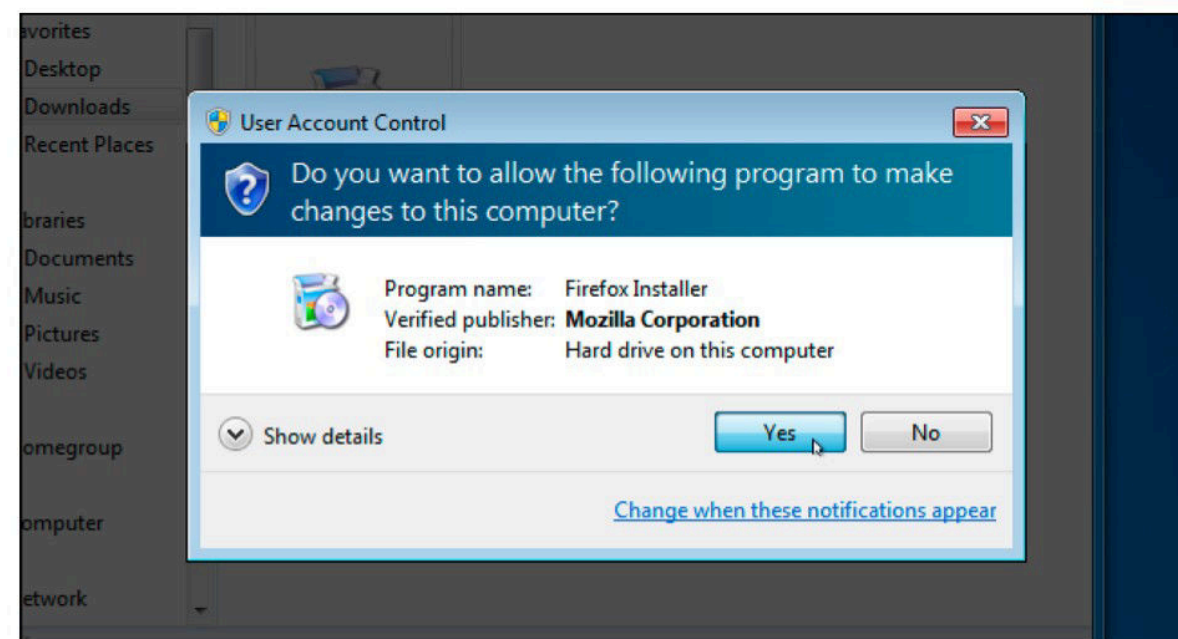
Blur



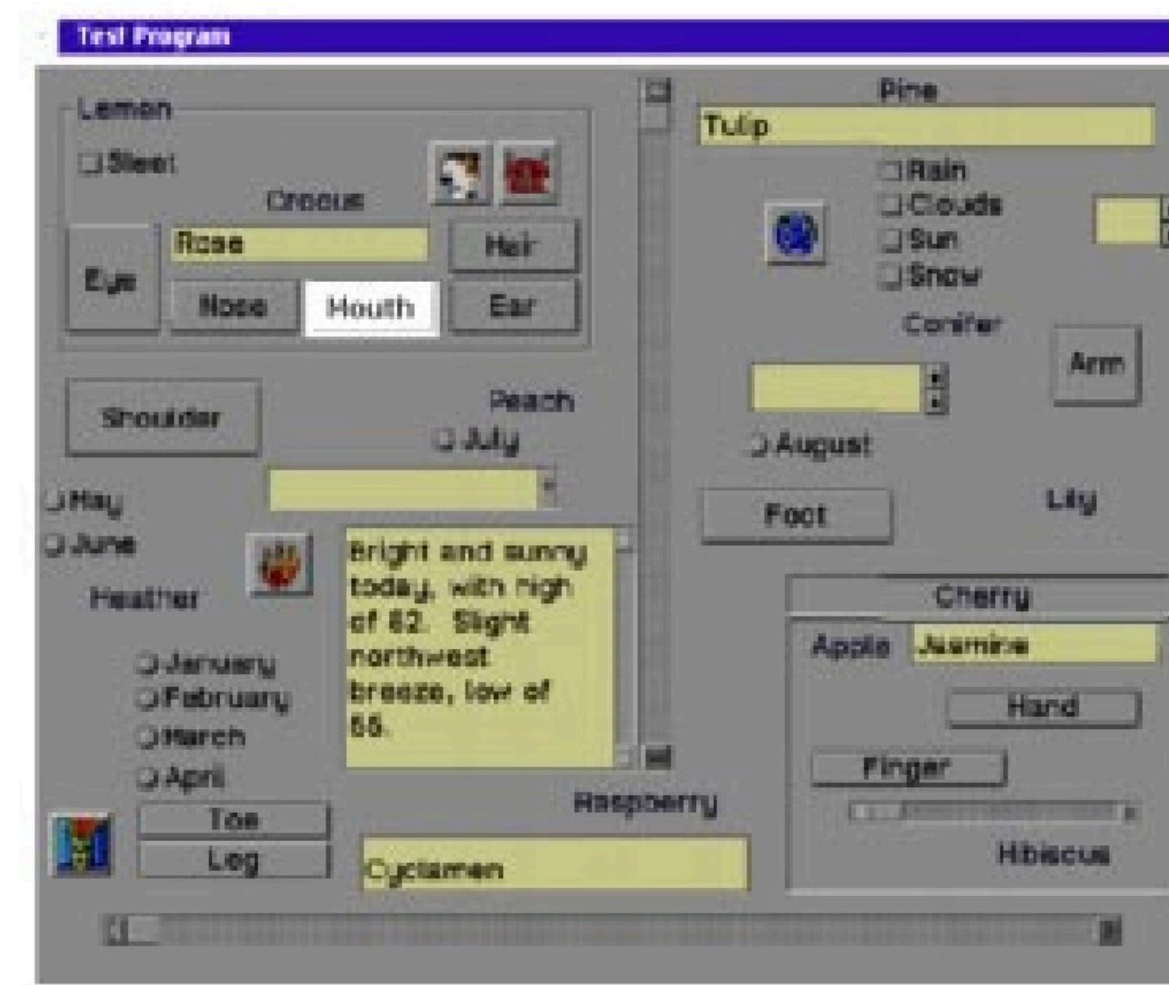
Darkening

Modulating Everything Else: Recommendation, Example

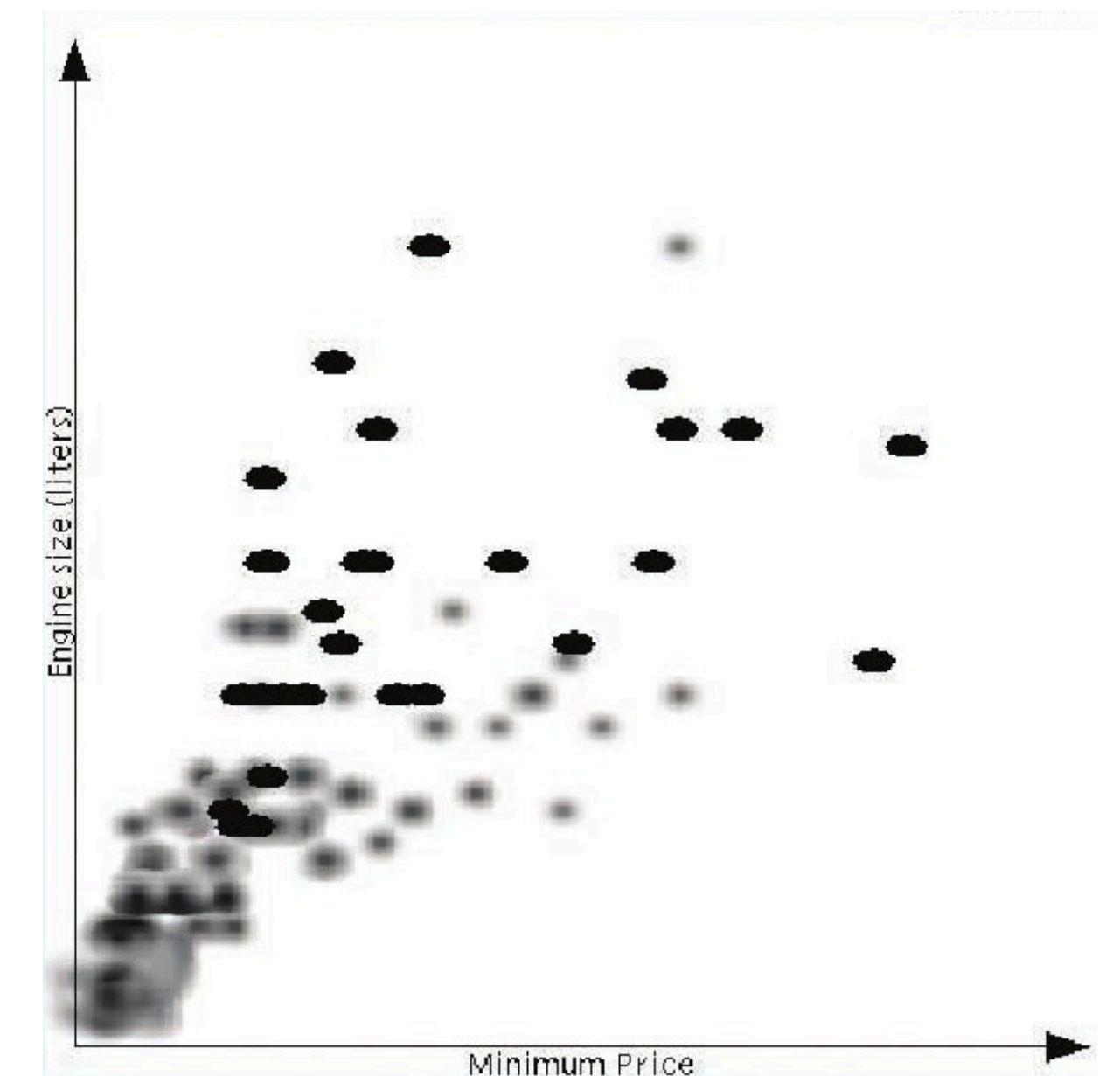
Don't use unless the **sole objective** is to guide attention toward one (set of) items



[MS Windows]



[Zhai et al., 1997]



[Kosara et al., 2002]

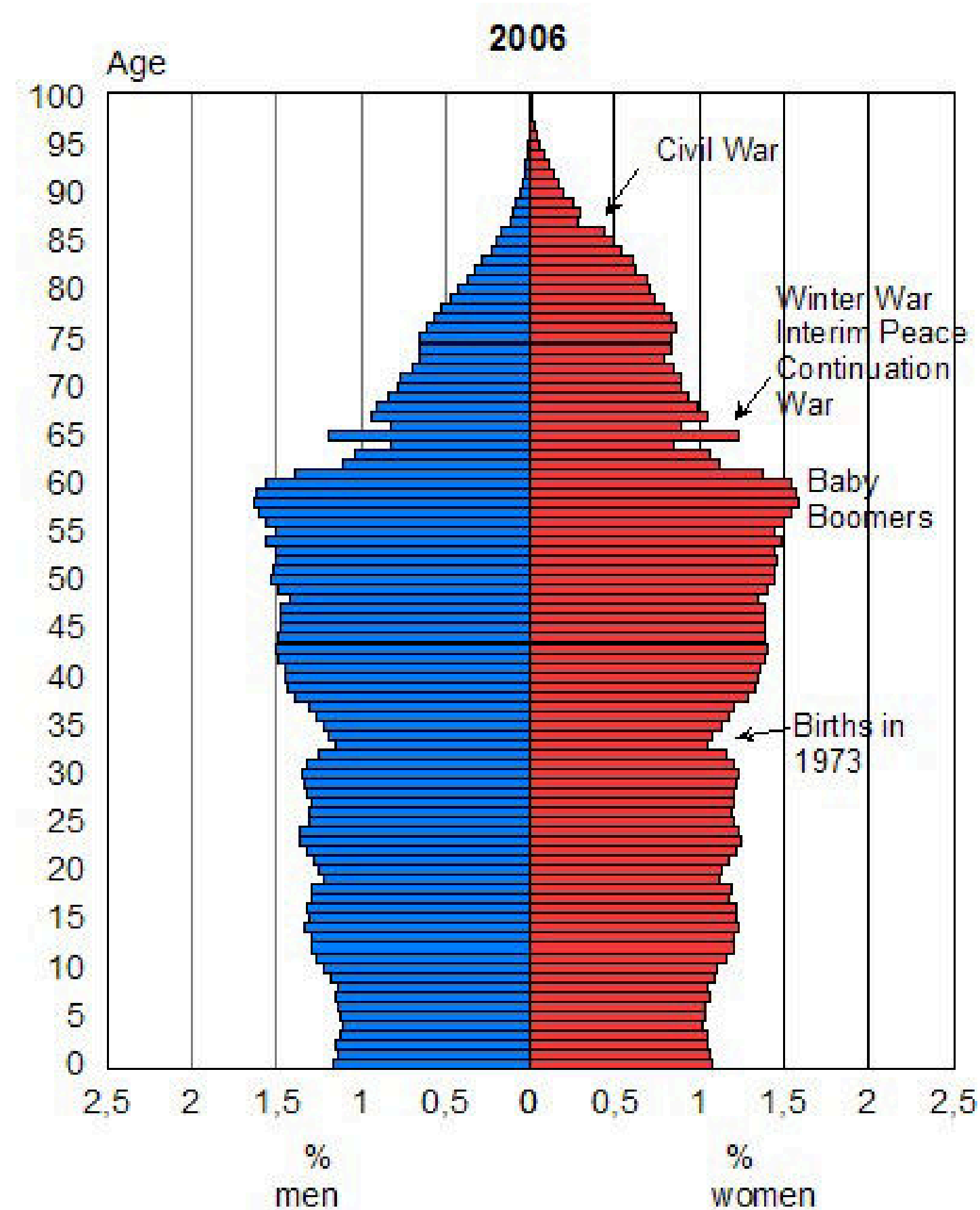
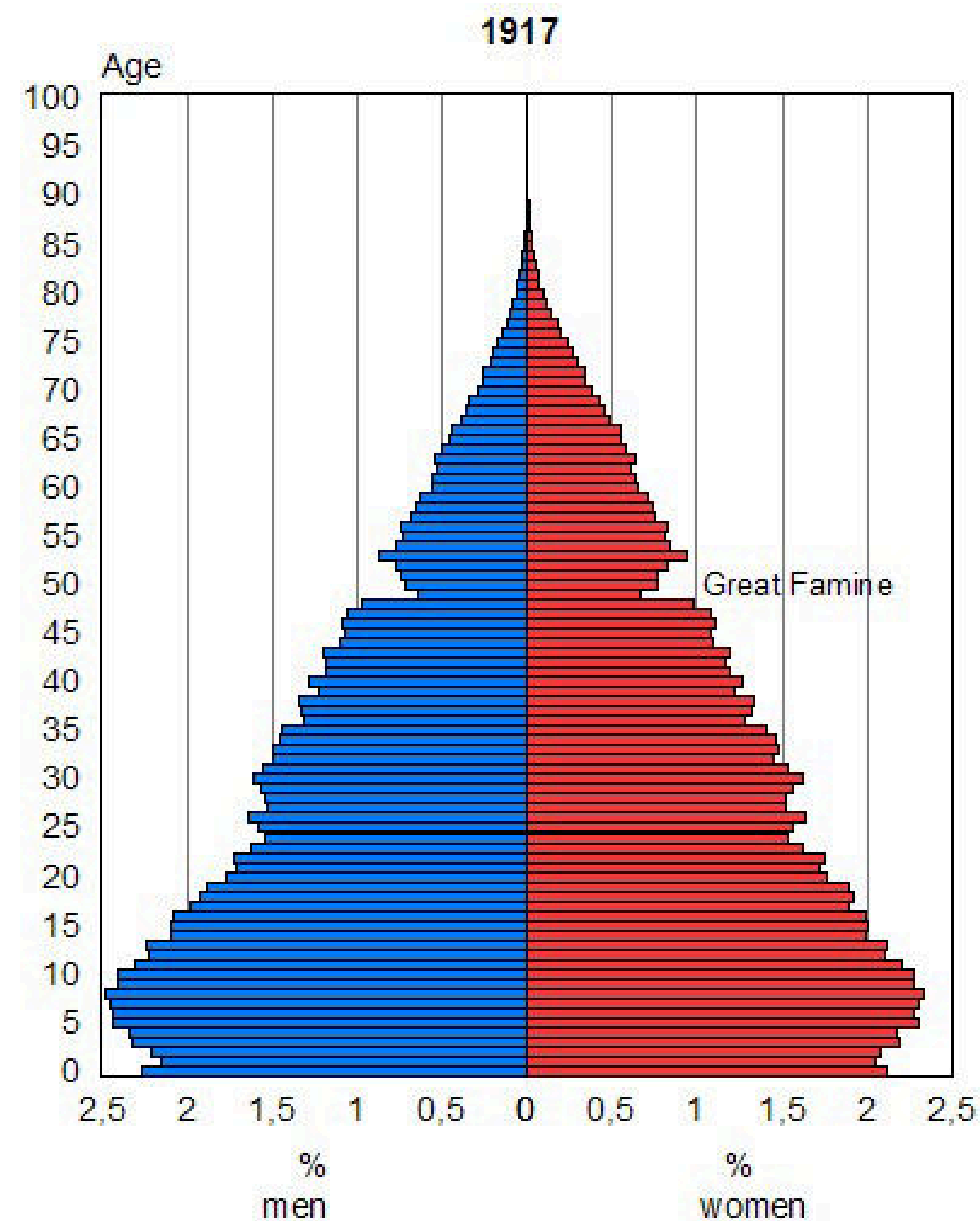
Gestalt Laws

- Proximity
- Similarity
- **Symmetry**
- Connectedness
- Continuity
- Closure
- Common Fate

When two symmetrical elements are unconnected the mind perceptually connects them to form a coherent shape

[] { } []

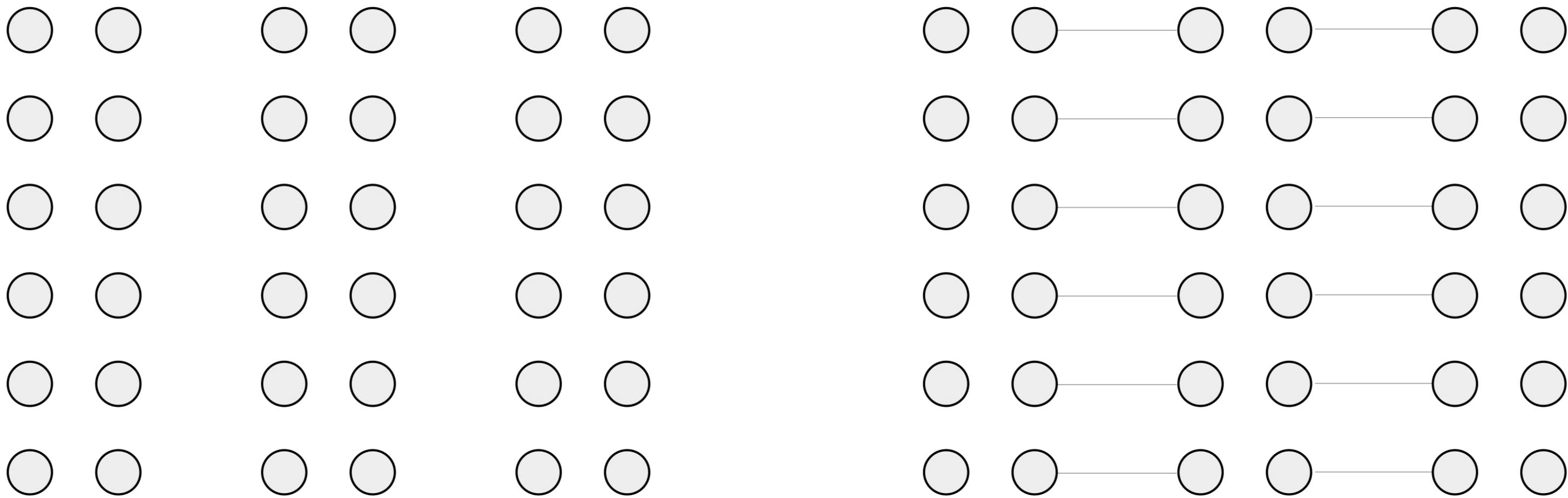
How does symmetry let you easily compare men vs women? What about asymmetry across plots?



Gestalt Laws

- Proximity
- Similarity
- Symmetry
- **Connectedness**
- Continuity
- Closure
- Common Fate

Elements that are visually connected are perceived as more related than elements with no connection

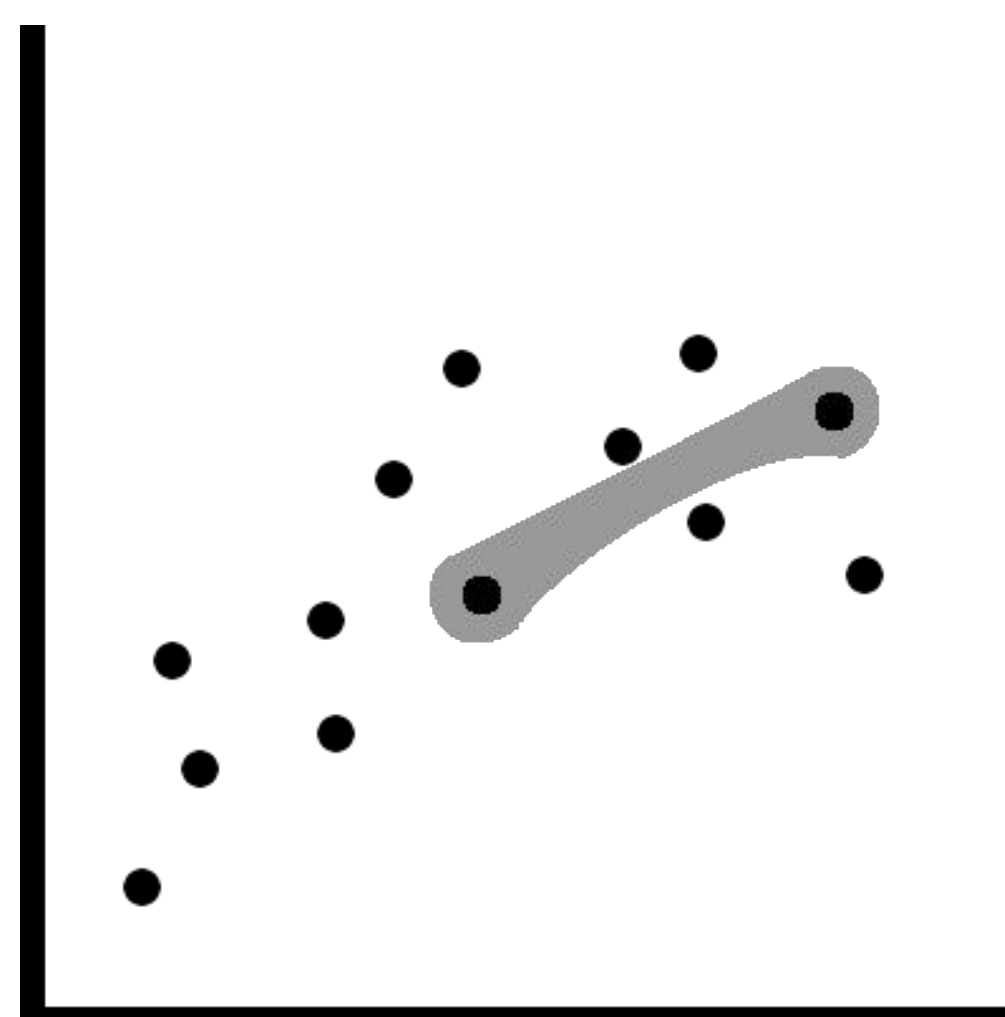


Connectedness overrides proximity

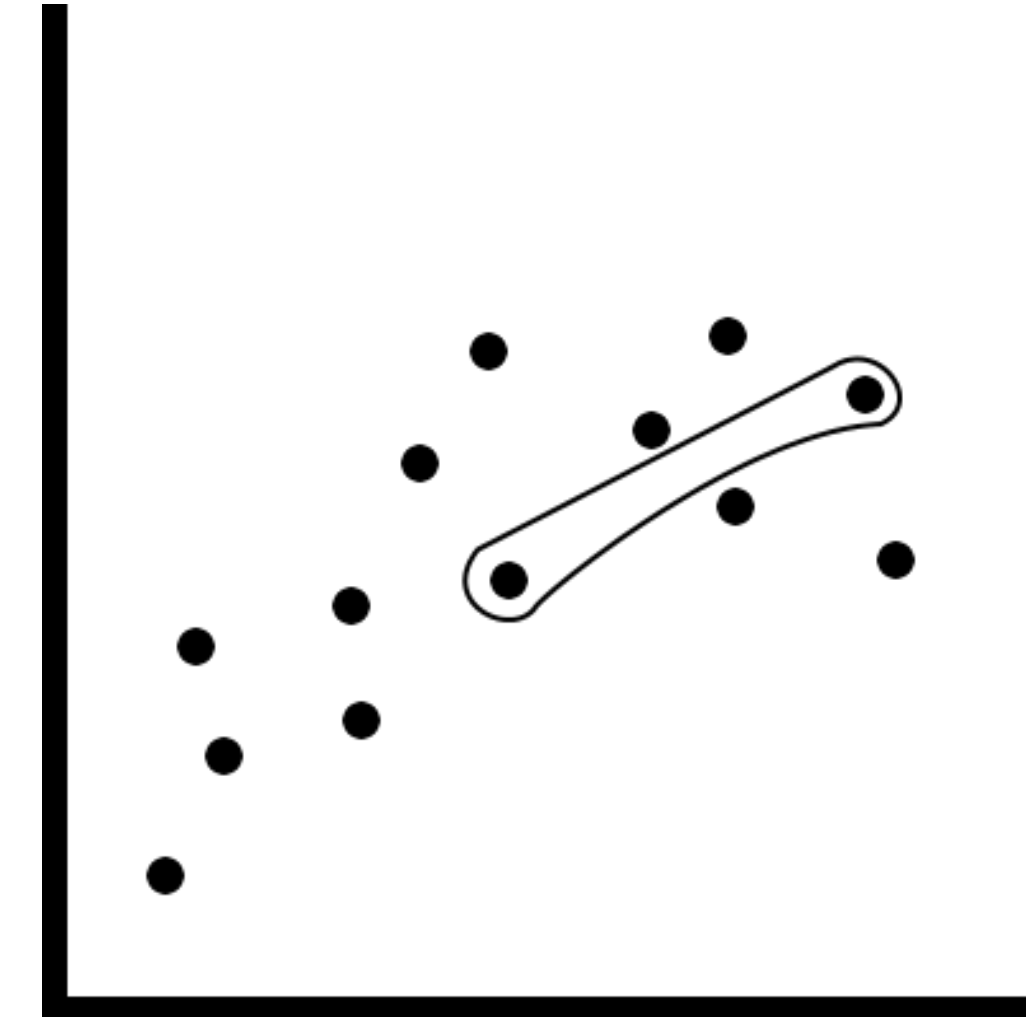
Connectedness



Connectedness



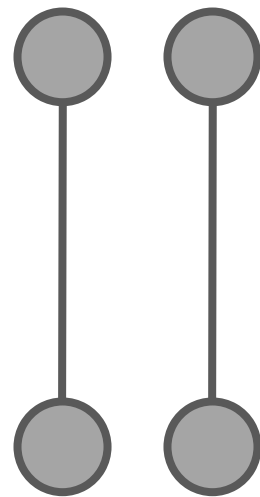
Surface



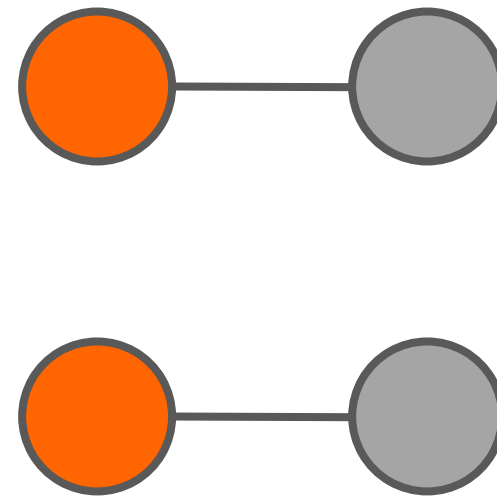
Outline

Common Region

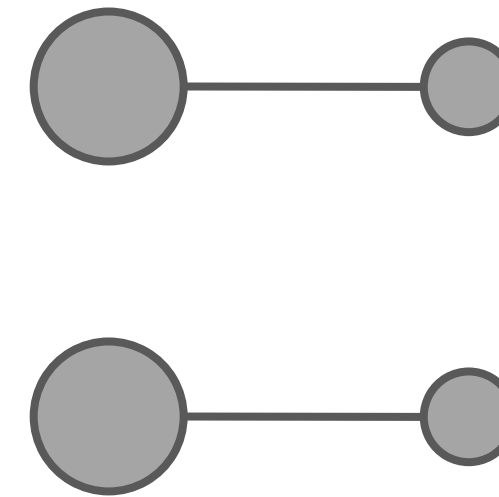
A Little Experiment...



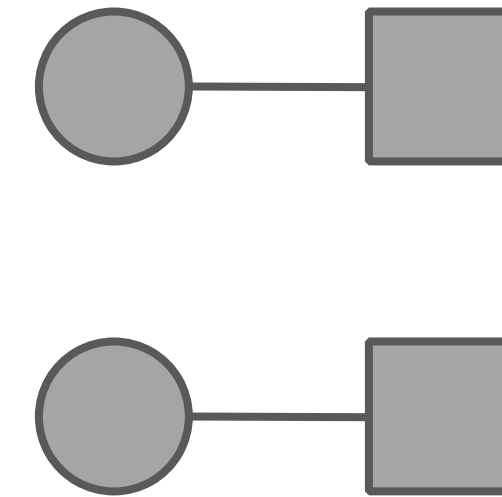
Proximity



Color



Size



Shape

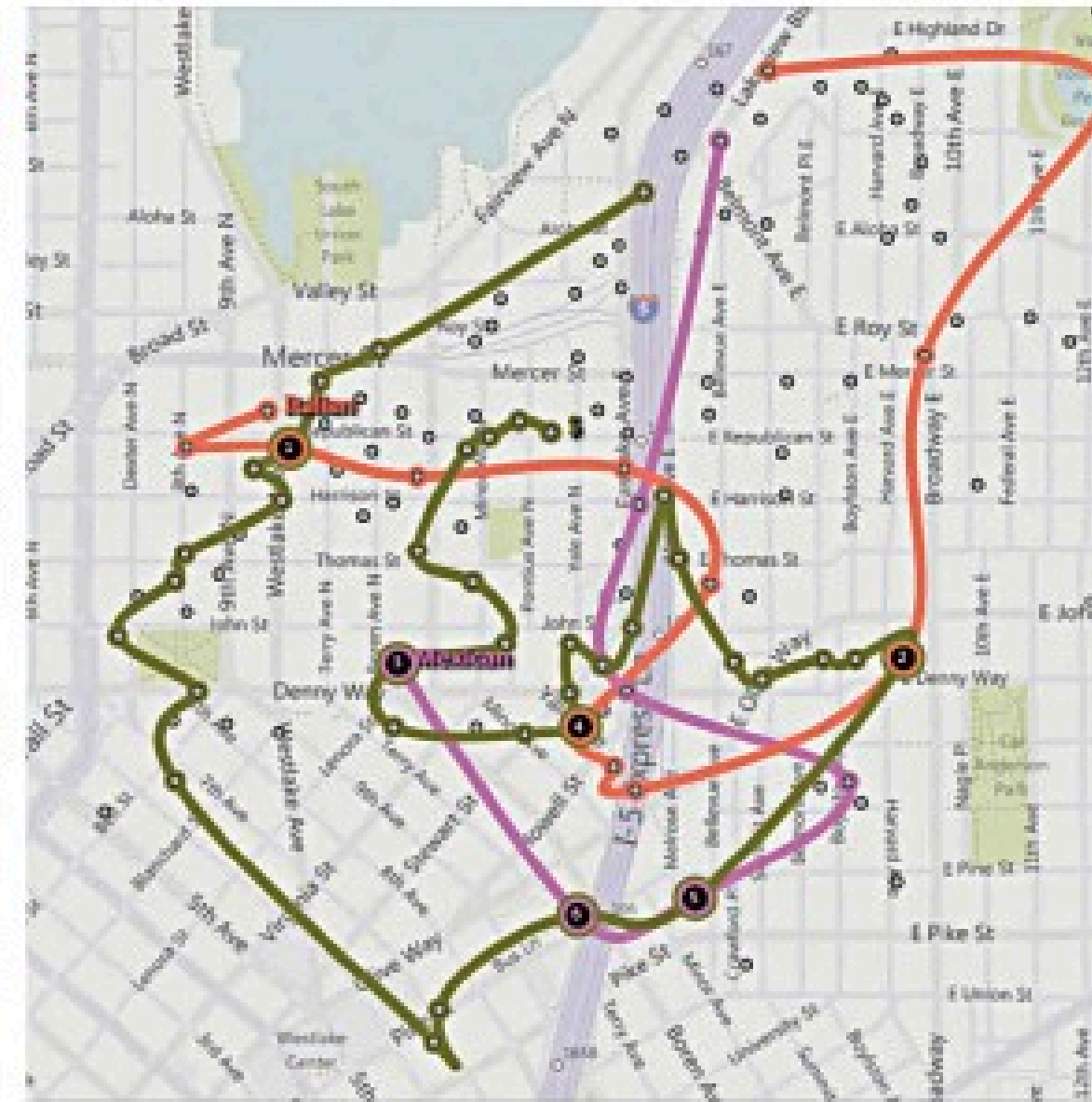
CONNECTEDNESS VARIETIES

Bubble Sets



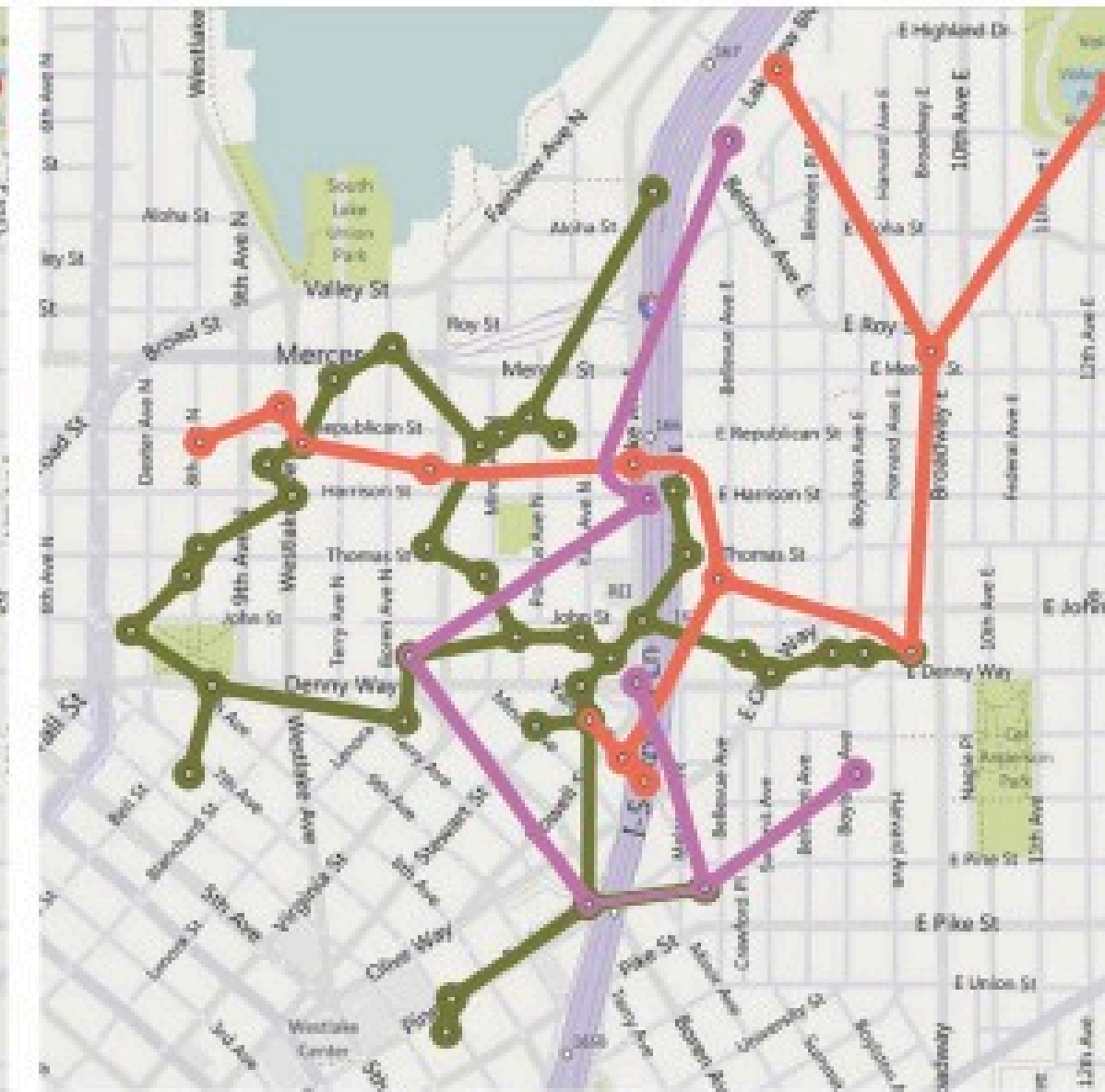
Image by [Dinkla et al., 2011]
Technique by [Collins et al., 2009]

Line Sets



[Alper et al., 2011]

Kelp Diagrams



[Dinkla et al., 2012]

Gestalt Laws

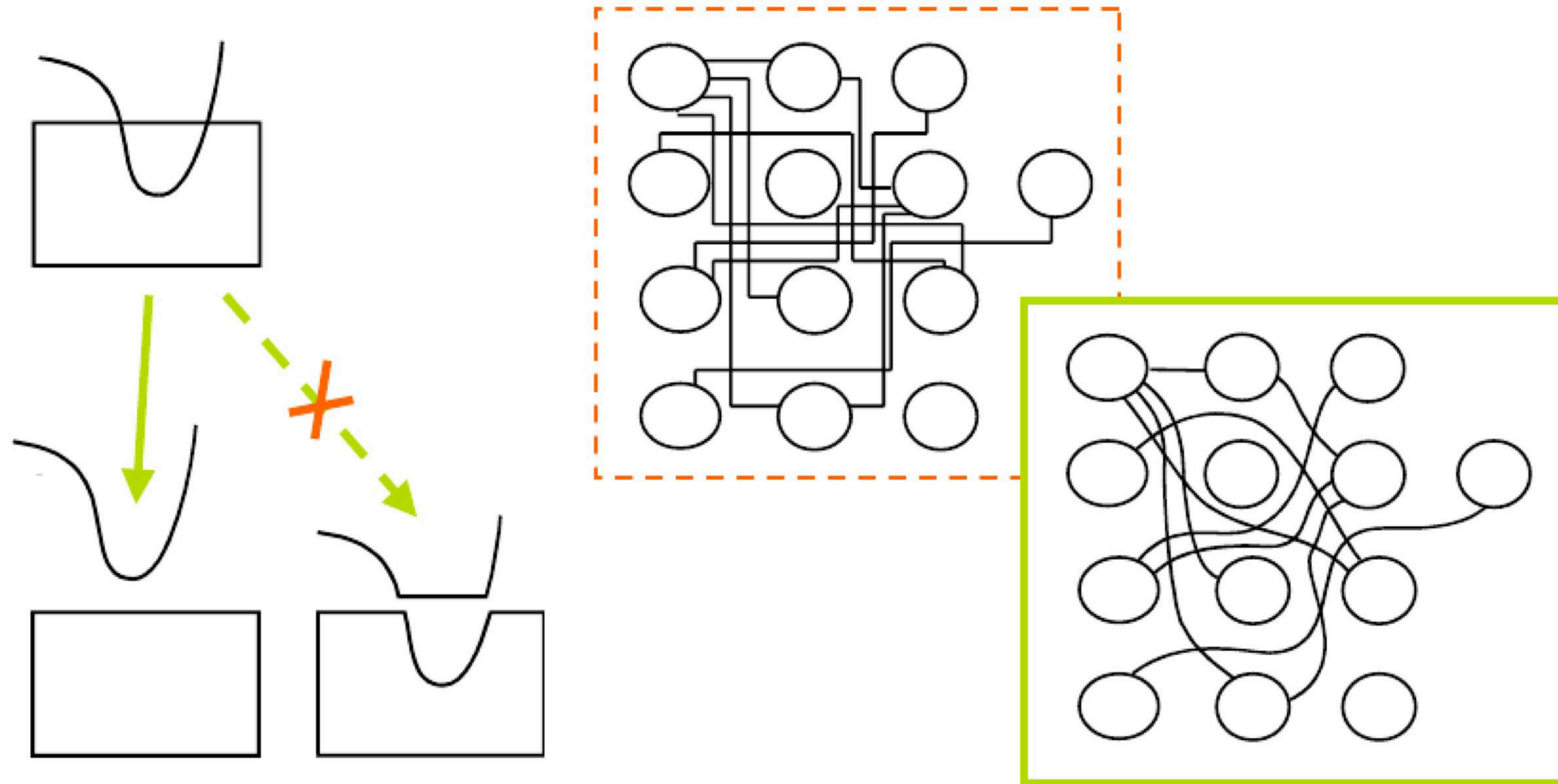
- Proximity
- Similarity
- Symmetry
- Connectedness
- **Continuity**
- Closure
- Common Fate

Human eye will follow the smoothest path when viewing lines, regardless of how the lines were actually drawn.

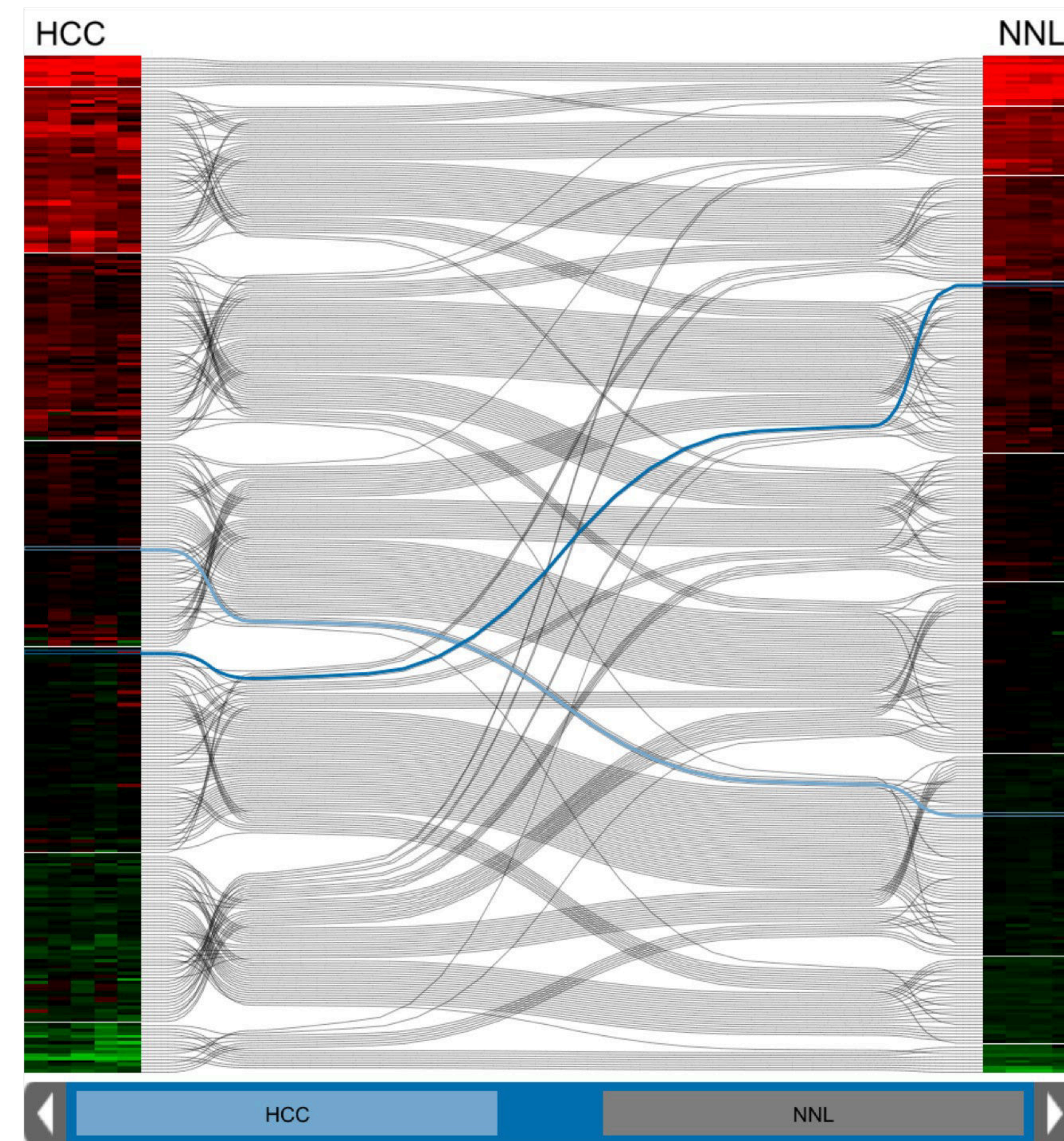
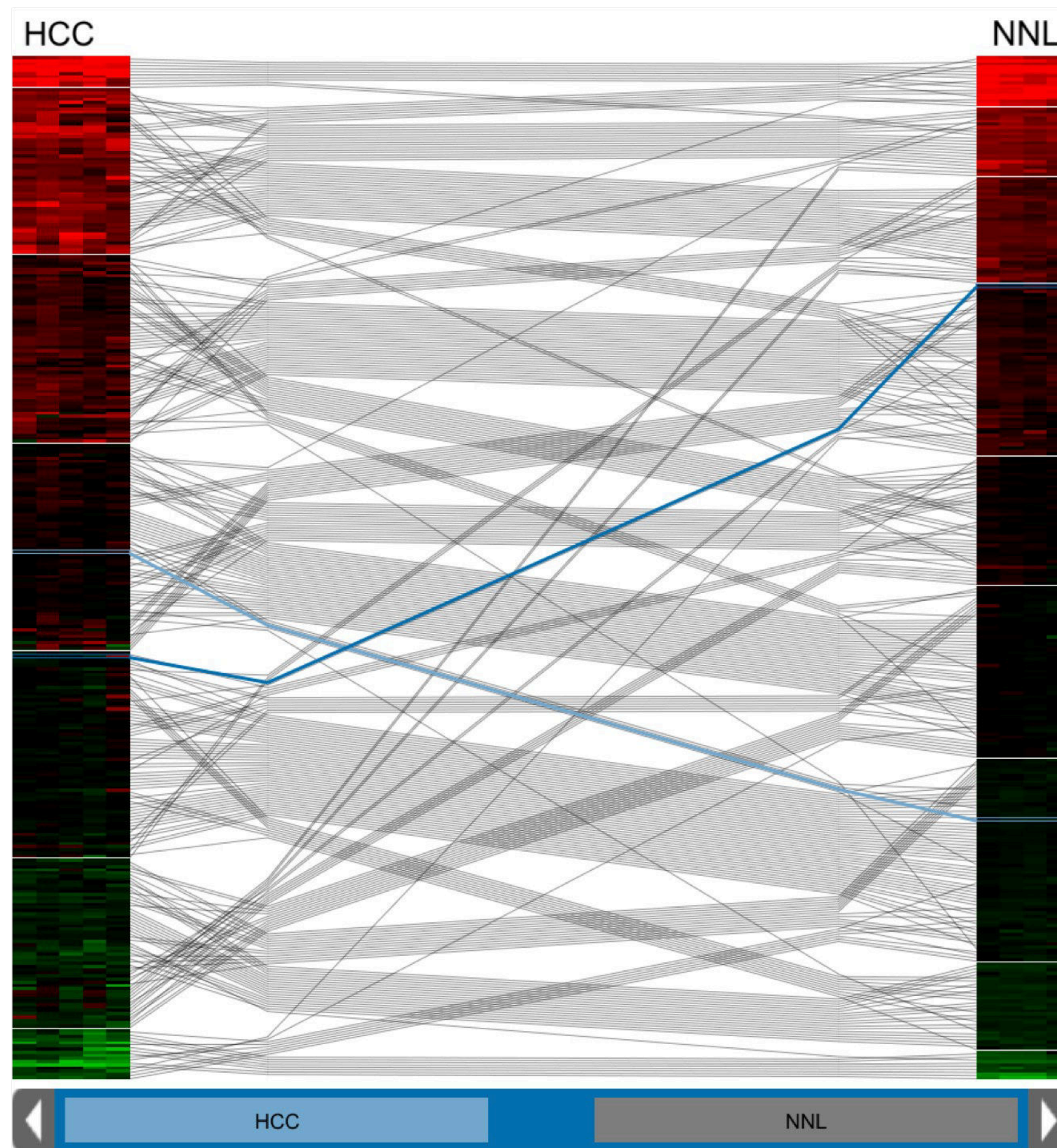


Continuity

Things: smooth & continuous



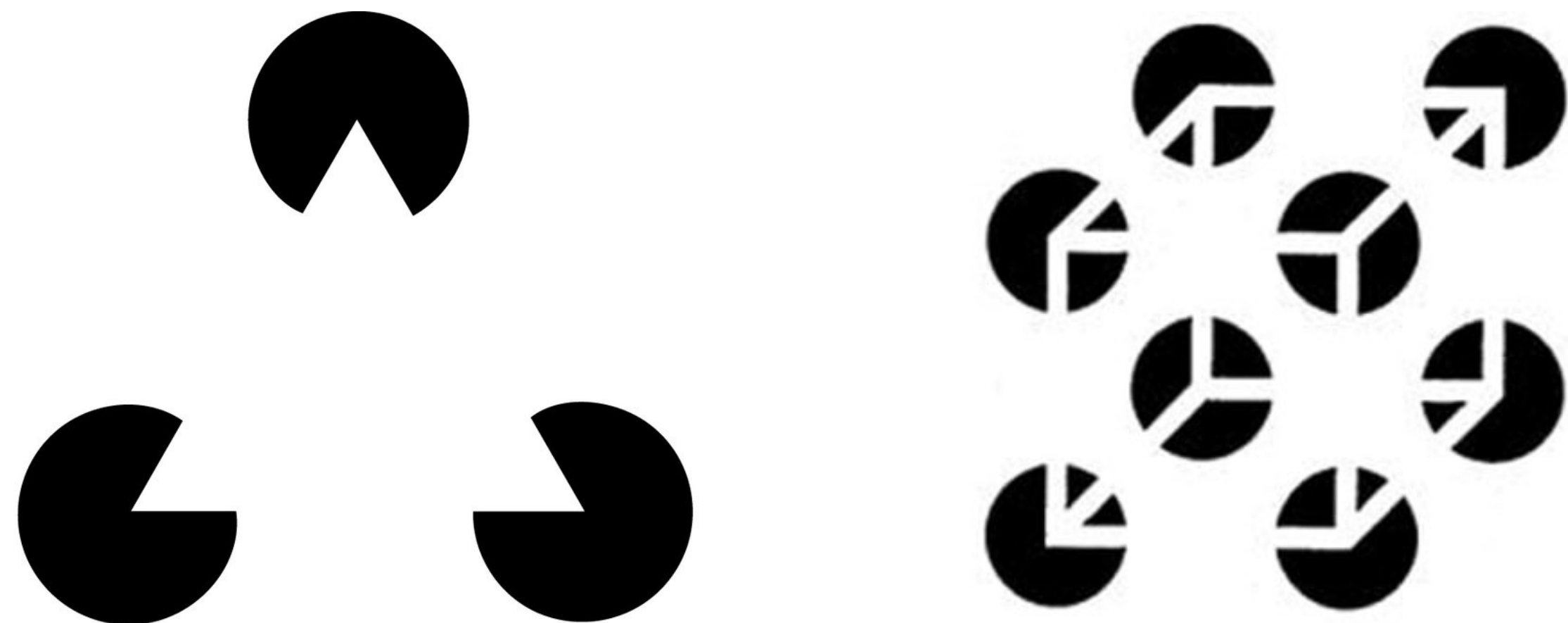
How does continuity make one of these plots easier to interpret?



Gestalt Laws

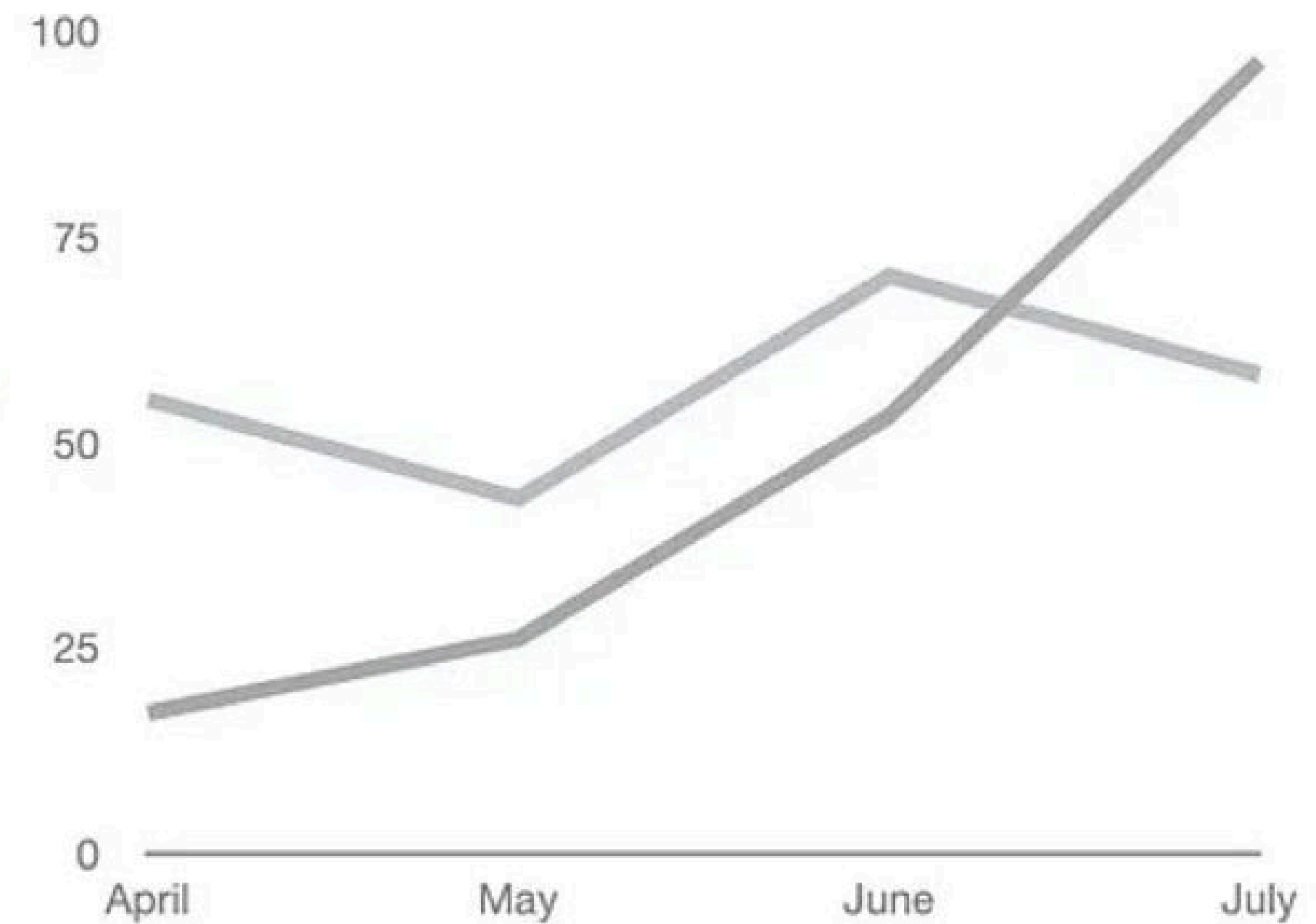
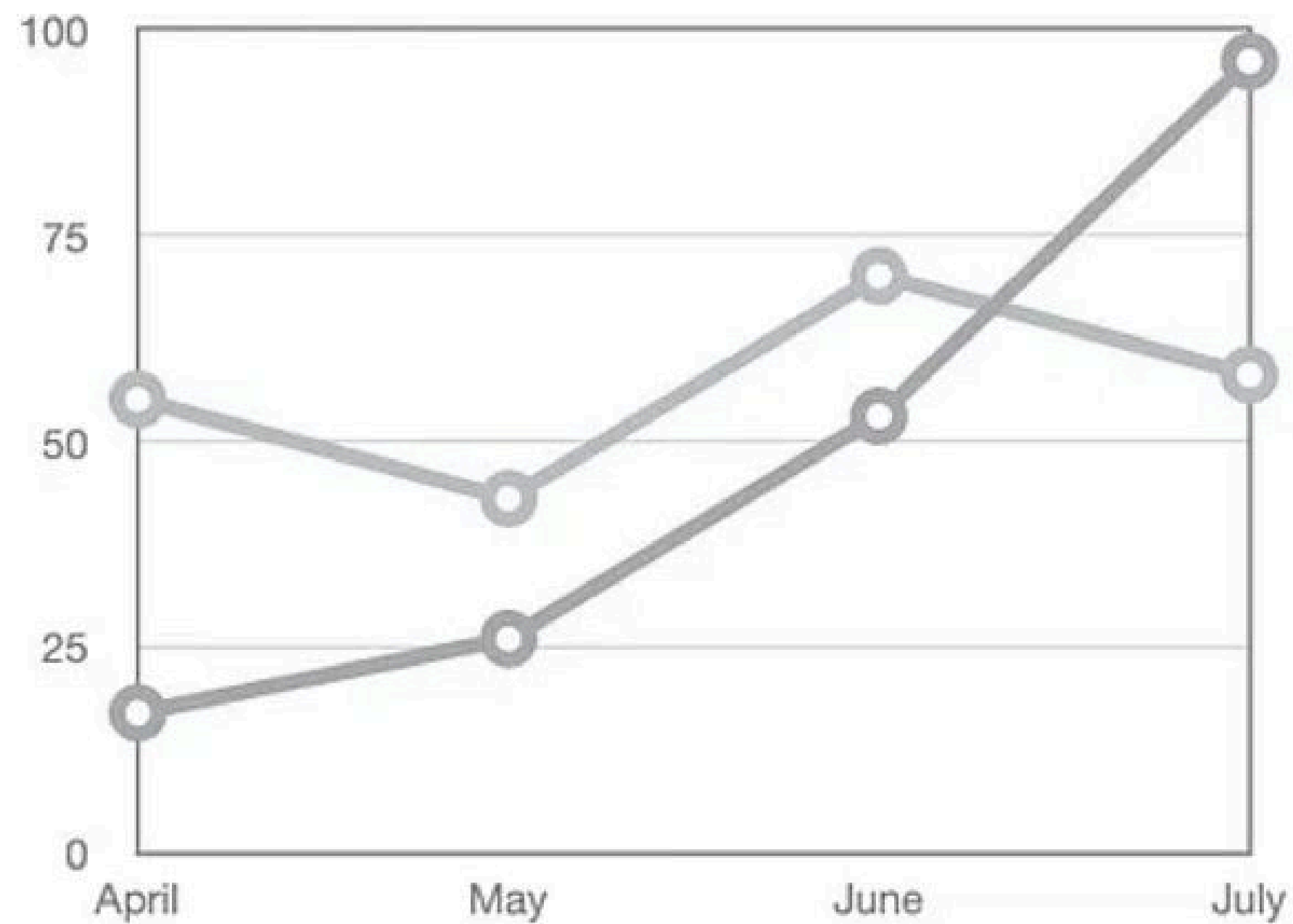
- Proximity
- Similarity
- Symmetry
- Connectedness
- Continuity
- Closure
- Common Fate

Humans tend to perceive objects as complete rather than focusing on the gaps that the object might contain.

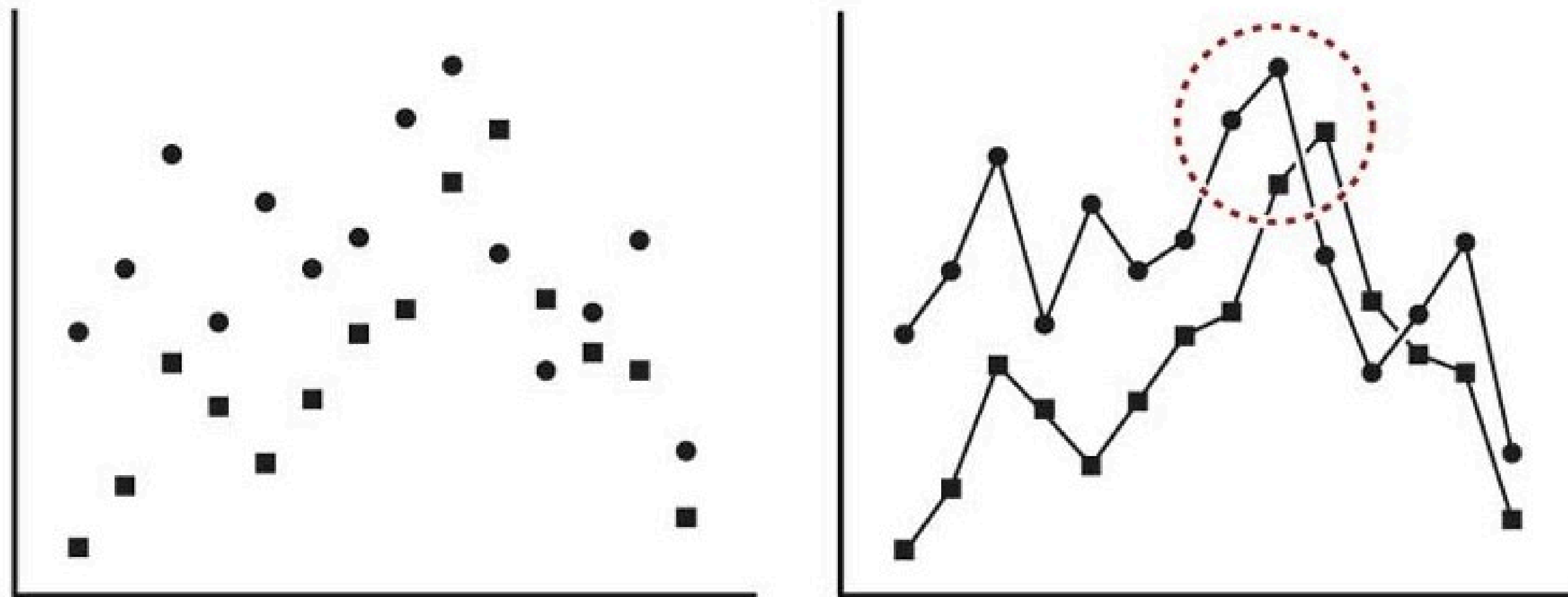


Our mind will automatically draw in the y axis, even if the lines to represent it aren't there

We can **remove “chartjunk”** in this way, because our mind will fill in what's missing. It must be something obvious to fill in (here, a straight line); this doesn't work with complex shapes.



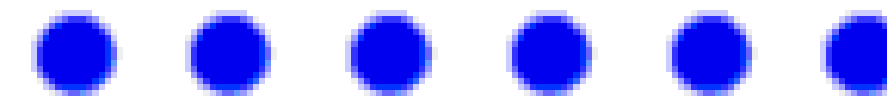
ENCLOSURE



Gestalt Laws

- Proximity
- Similarity
- Symmetry
- Connectedness
- Continuity
- Closure
- Common Fate

Humans tend to perceive elements moving in the same direction as being more related than elements that are stationary or that move in different directions.



Continuity and Common Fate: an example

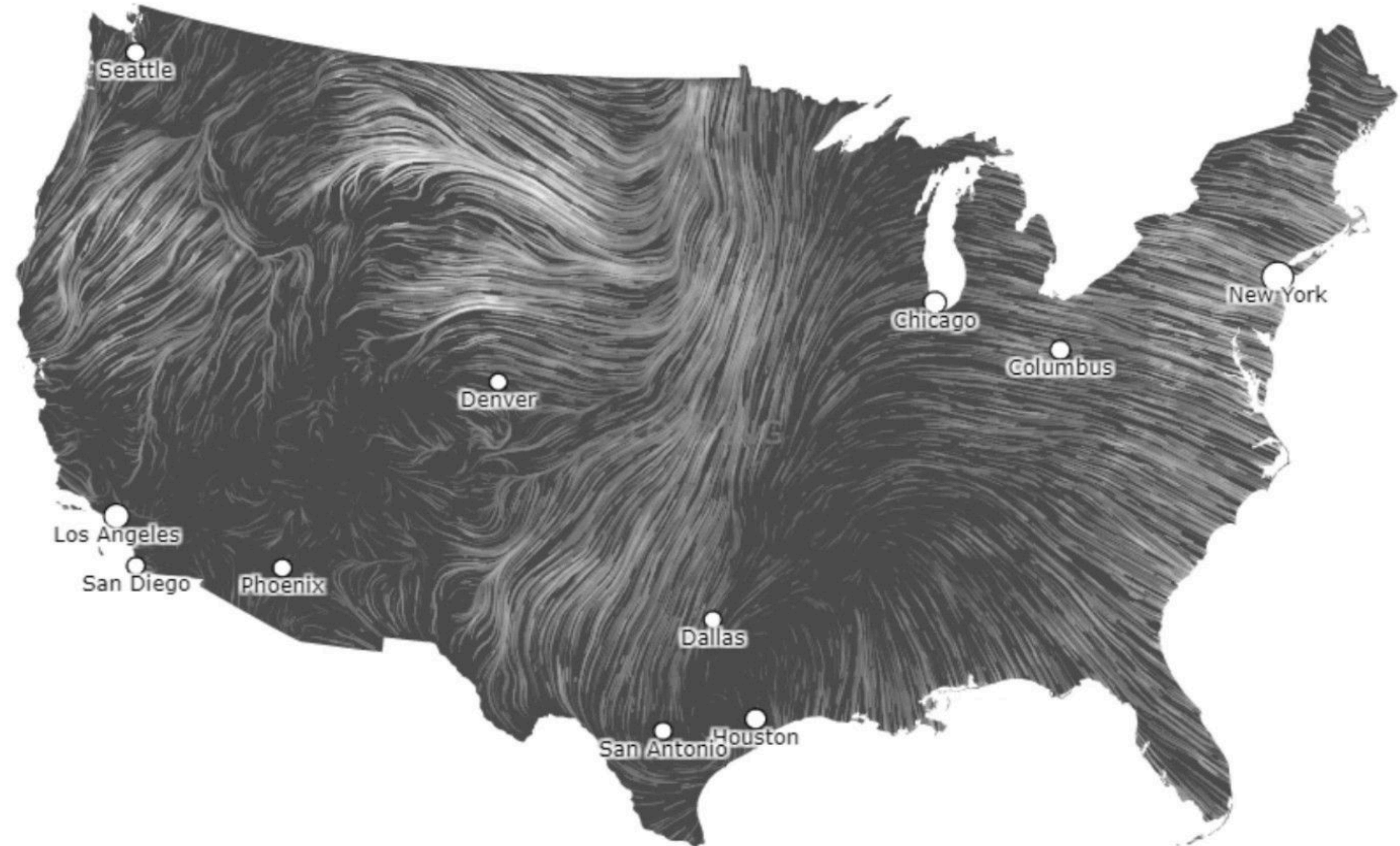
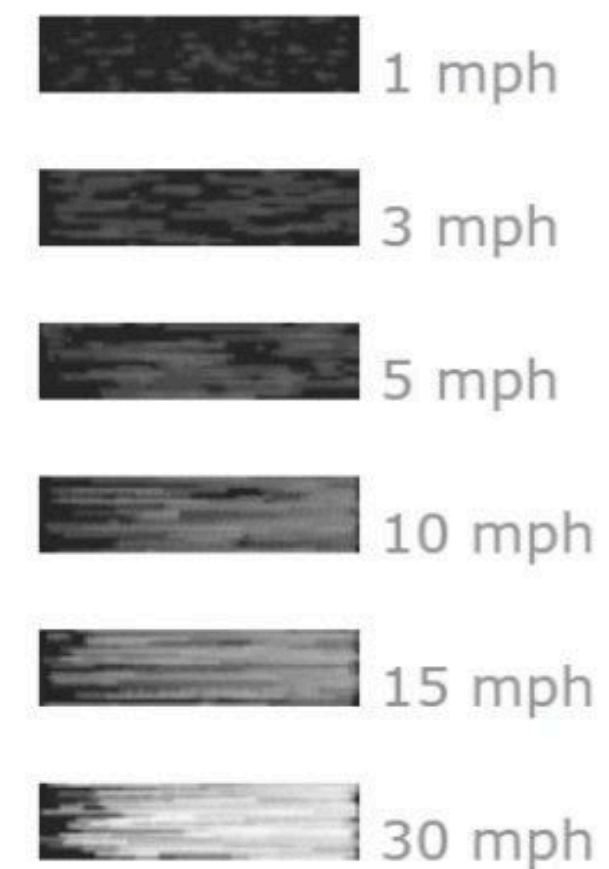
March 12, 2022

5:12 pm EST

(time of forecast download)

top speed: **50.5 mph**

average: **14.4 mph**



Relative versus
absolute differences

Which is longer?



Weber's Law

Which is longer?

0.97"



1.17"



Which is longer?



Which is longer?

3.97"



4.17"



Weber's Law:

we judge based on relative, not absolute differences

Just Noticeable Difference $\rightarrow$ $\frac{\Delta I}{I} = k$ $\leftarrow$ Constant, empirically determined,
different for different types of stimulus

$\uparrow$
Original stimulus intensity

In order for a difference to be noticeable, the amount something must be changed (JND) is a fixed proportion of the reference sensory level

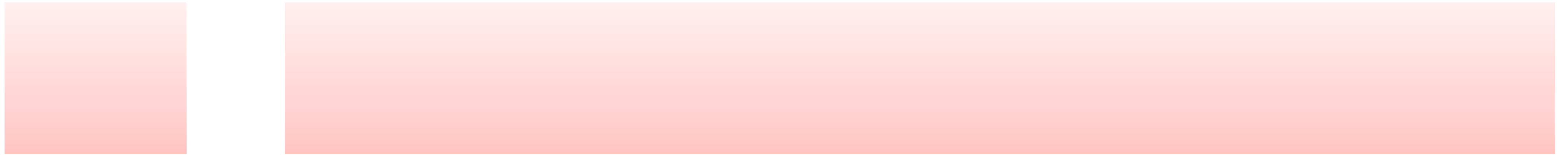
A good approximation for some sensory dimensions, e.g. brightness

How much bigger?

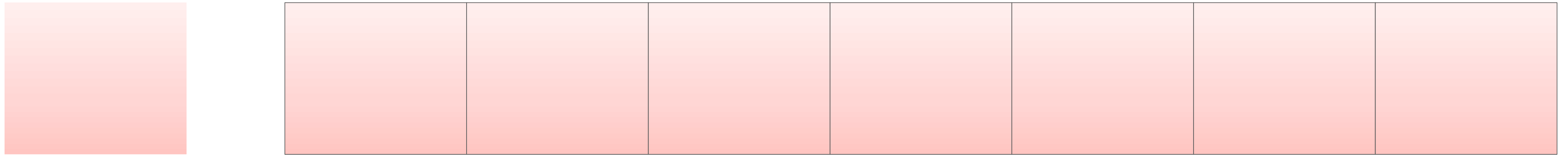
Steven's Power Law

How much bigger?

How much bigger?



How much bigger?



We perceive different visual channels with different levels of accuracy.

Steven's Power Law

Perceived Magnitude

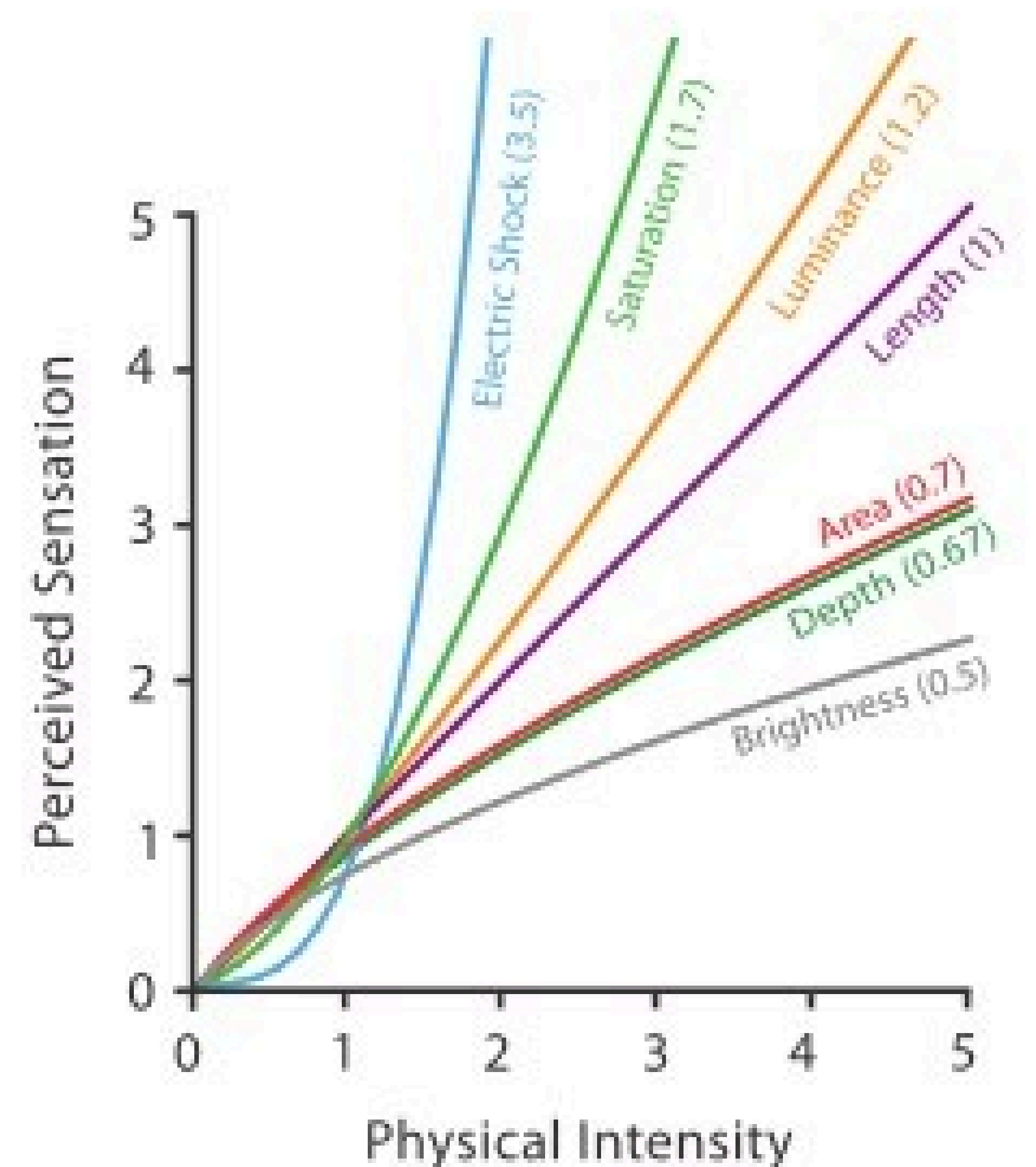
$$S = I^a$$

Exponent,
empirically determined

Magnitude in physical units

We don't accurately perceive changes in area; increasing area X4 only is perceived as a 3X increase

We are pretty accurate at determining changes in length



In class activity

- Once again, we are going to think about how these concepts can be used when building our own charts
- Take a look at your assigned chart and concept. Using that concept, come up with a suggestion for how to improve the chart.
- Be prepared to share your suggestion with the class!

Group	Concept
A	Proximity
B	Similarity
C	Symmetry
D	Connectedness
E	Continuity

Group	Concept
F	Closure
G	Weber’s law
H	Steven’s power law
I	Discriminability
J	Separability
K	Common Fate